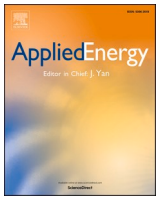
Applied Energy 408 (2026) 127369



Contents lists available at ScienceDirect

Applied Energy

journal homepage: www.elsevier.com/locate/apenergy



Bi-level planning of data centers with coupled electricity-heat-computation system using data-driven scenario generation for representing uncertainties☆

Lingfang Yang [a], Yujie Lin [b], Mohammad Shahidehpour [c], Yuanyi Chen [d], Qiang Yang [a,*]

[a] College of Electrical Engineering, Zhejiang University, Hangzhou 310027, China
[b] Dispatching and Control Center of China Southern Power Grid, Guangzhou, Guangdong 510663, China
[c] Department of Electrical and Computer Engineering, Illinois Institute of Technology, Chicago 60616, USA
[d] School of Electrical and Automation Engineering, Nanjing Normal University, Nanjing 210023, China

HIGHLIGHTS

- A diffusion model based operational scenario generation method is presented.
- An CEHCS expansion planning model is formulated to upgrade the DTC.
- Safe operating temperature of CPUs is considered in constraints.
- A heuristic bi-level scheme is designed to solve the nonlinear planning model.
- Chance constraints of energy storage systems are adopted to cope with dispatch risks.

ARTICLE INFO

Keywords:
data center
diffusion model
scenario generation
heuristic bi-level scheme
electricity-heat-computation coupling

ABSTRACT

Along with the rapidly increasing data traffic, data centers (DTCs) have considered the expansion of their server capacity, which has led to higher energy consumption. To deal with this issue, an effective solution is presented for the coupled electricity-heat-computation system (CEHCS) in DTCs. However, CEHCS uncertainties of loads and energy resources can pose additional challenges for the optimal planning of DTC. Therefore, this work proposes a bi-level expansion planning of DTCs, which considers data-driven scenario generation for representing CEHCS uncertainties. First, a diffusion model is designed using the operational scenario generation (DMOSG) method to characterize the multi-dimensional randomness of multivariate time series, where a one-dimensional U-net works as the denoising network with an interpretable latent output space. Then, a DTC expansion planning model with CEHCS is formulated considering the safe operation temperature of CPUs, where the operational status of each server is individually characterized to increase the proposed model's fidelity. Although the DMOSG helps address the randomness of input scenarios for planning, it is also necessary to consider the uncertainty hedging for energy scheduling. Thus, chance constraints are included for BESSs to cope with risks during CEHCS dispatch. A heuristic solution is proposed for the bi-level scheme to solve the DTC planning model. At the upper level, the number of servers to be added is determined by the particle swarm optimization (PSO) algorithm. The upper-level solution is submitted to the lower level, where the performance cost of DTC planning and scheduling decisions is obtained by the Gurobi solver. Then, the DTC performance cost is returned to the upper level iteratively for calculating the optimal DTC planning results. The proposed DTC planning solution with CEHCS is validated through case studies, where the numerical results confirm the accuracy of the proposed characterization of operational uncertainty, as well as the cost-effectiveness and improved energy efficiency of the DTC expansion planning scheme.

☆ This article is part of a Special issue entitled: 'Low-Carbon IES(Jiehui Zheng)' published in Applied Energy.

* Corresponding author.
E-mail address: qyang@zju.edu.cn (Q. Yang).

https://doi.org/10.1016/j.apenergy.2026.127369
Received 4 September 2025; Received in revised form 30 November 2025; Accepted 5 January 2026
Available online 14 January 2026
0306-2619/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Nomenclature

Abbreviation

DTC	Data center
GDC	Green data center
IT	Information technology
CEHCS	Coupled Electricity-heat-computation system
DMOSG	Diffusion model based operational scenario generation
KSF	Kernel smoothing function
PUE	Power usage effectiveness
WHRS	Waste heat recovery system
ISP	Internet service provider
PV	Photovoltaic
WT	Wind turbine
DHS	District heating system
CHP	Combined heat and power
BESS	Battery energy storage system
HSS	Heating storage system
RES	Renewable energy source
MCS	Monte Carlo sampling
LHS	Latin hypercube sampling
KSF	Kernel smoothing function
CAPEX	Capital expenditure
OPEX	Operating expense
PSO	Particle swarm optimization

Indices and Sets

t	Index of time slots
i	Index of representative days
j	Index of servers
n	Index of buses in power systems
k	Index of particles in PSO
$iter$	Index of iterations in PSO based bi-level scheme
Ω^{ge}	Set of representative scenarios clustered from generated scenarios
Ω^{real}	Set of historical scenarios
Ω_n^{source}	Set of sources located at the bus n
Ω_n^{branch}	Set of branches connected to the bus n

Parameters

T	Number of time slots in one day
P^{idle}	Idle power consumption of a server (kW)
P^{peak}	Peak power consumption of a server (kW)
β_1/β_2	Empirical coefficients of cooling systems
$Q^{cool,exist}$	Existing capacity of cooling systems (kW)
$T_{i,t}^{out}$	Temperature of outdoor air (°C)
$\mu_{i,t}$	Mean utilization rate of servers
R_i	Equivalent heat resistance (°C/kW)
C^{DC}	Heat capacity of the data center (kWh/°C)
θ^{IT}	Heat dissipation rate of IT equipment
$T^{in,min}/T^{in,max}$	Minimum/maximum indoor temperature (°C)
$r^{in,min}/r^{in,max}$	Minimum/maximum changing rate of indoor temperature (°C/h)
$\lambda_{i,t}^{inter}$	Arrival rate of interactive requests
$\lambda_{i,\tau}^{batch,total}$	Total number of batch requests that arrive at time τ
t_τ^{ddl}	Deadline of batch requests that arrive at time τ
$t_\tau^{delay,max}$	Maximum allowed delay time for the batch requests arrived at time τ
$\lambda^{srv,max}$	Maximum processing rate of servers
t^{trans}	Transmission delay (h)
$t^{inter,max}$	Maximum allowed delay time for interactive requests (h)
θ^{whrs}	Electricity to heat ratio of WHRSs
ω^{re}	Waste heat recovery efficiency
c_{equip}^{o}	Unit operation cost of equipment (CNY/kWh)
$c^{f,chp}$	Unit fuel cost of CHPs (CNY/kWh)
c^{dhs}	Unit price of heat purchased from DHS (CNY/kWh)
$c_{i,t}^{b}/c_{i,t}^{s}$	Price of buying/selling electricity (CNY/kWh)
$\rho^{g}/\rho^{dhs}/\rho^{chp}$	Emission factor of power grid/GBs/CHPs (g/kWh)
$P_{i,t}^{pv,max}$	Maximum power generation of PVs (kW)
$P_{i,t}^{wt,max}$	Maximum power generation of WTs (kW)
ϖ	Ramping rate of CHPs (kW/h)
ω^{chp}	Electric-heating ratio of CHPs
Soc^{min}/Soc^{max}	Minimum/maximum state of charge of BESSs (kWh)
λ^{es}	Self-loss rate of BESSs
$\vartheta^{es,c}/\vartheta^{es,d}$	Charge/discharge efficiency
$P_t^{es,c,max}/P_t^{es,d,max}$	Maximum charging/discharging power of BESSs (kW)
Soh^{min}/Soh^{max}	Minimum/maximum heat storage state of HSSs (kWh)
λ^{hs}	Self-loss rate of HSSs
$\vartheta^{hs,c}/\vartheta^{hs,d}$	Heat absorption/release efficiency
$H_t^{hs,c,max}/H_t^{hs,d,max}$	Maximum heat absorption/release power of HSSs (kW)
ω	Velocity coefficient in PSO
c_1/c_2	Self-learning factor/global learning factor
r_1/r_2	Random numbers
$x_{j,k}$	Reactance of the branch between buses j and k

Variables and Functions

$P_{i,t}^{DC}$	Electric power consumed by the DTC (kW)
$P_{i,t}^{IT}$	Electric power consumed by IT equipment (kW)
$P_{i,j,t}^{srv}$	Electric power consumed by a server (kW)
$P_{i,t}^{cool}$	Electric power consumed by cooling systems (kW)
$P_{i,t}^{whrs}$	Electric power consumed by WHRSs (kW)
$H_{i,t}^{cool}$	Heat extracted by cooling systems from indoor air (kW)
$H_{i,t}^{IT}$	Heat dissipation power of servers (kW)
$T_{i,t}^{in}$	Temperature of indoor air (°C)
$r_{i,t}^{in}$	Changing rate of indoor temperature (°C/h)
$\lambda_{i,\tau,t}^{batch}$	Number of batch requests that arrive at time τ and are processed at time t
$\lambda_{i,t}^{batch}$	Total number of batch workloads being processed at time t
$H_{i,t}^{re}$	Recovered heating power (kW)
F	Average scheduling costs at the lower level (CNY)
F_i^{o}	Operation cost (CNY)
F_i^{fuel}	Fuel cost (CNY)
F_i^{grid}	Electricity transaction cost with the utility grid (CNY)
F_i^{carbon}	Carbon emission cost (CNY)
F_{equip}^{om}	Operation and maintenance cost of equipment(CNY)
$P_{i,t}^{pv}$	Electric power generation of PVs (kW)
$P_{i,t}^{wt}$	Electric power generation of WTs (kW)
$H_{i,t}^{dhs}$	Heating power purchased from the DHS (kW)
$H_{i,t}^{chp}$	Heating power of CHPs (kW)
$P_{i,t}^{chp}$	Electric power generation of CHPs (kW)
$P_{i,t}^{es,c}/P_{i,t}^{es,d}$	Charging/discharging power of BESSs (kW)
$H_{i,t}^{hs,c}/H_{i,t}^{hs,d}$	Power for absorbing/releasing heat of HSSs (kW)
$P_{i,t}^{g,b}$	Electrical power purchased from the utility grid (kW)
$H_{i,t}^{load}$	Heating load (kW)
$Soc_{i,t}$	State of charge of BESSs in the ith DTC at time t (kWh)
$\eta_{i,t}^{es,c}/\eta_{i,t}^{es,d}$	Charging/discharging state of BESSs

$Soh_{i,t}$	Heat storage of HSSs in the ith DTC at time t (kWh)	$fit(\cdot)$	Fitness value of particles, i.e. performance cost
$\eta_{i,t}^{hs,c}/\eta_{i,t}^{hs,d}$	Heat absorption/release state of HSSs	θ_j/θ_k	Phase angle of buses j and k
V_k^{iter}	Velocity of particles	$u_{n,l}$	Binary variable indicating the direction of power flow
$pBest_k$	Position where particle k reflects its lowest fitness value	$P_{i,t,n,s}$	Electric power of the source
$gBest$	Position with the lowest fitness value across all particles	$P_{i,t,n,l}$	Power flow of branch l that is connected to bus n
Q_k^{iter}	Position of particles	$P_{i,t,n}^{load}$	Electricity demand of bus n

1. Introduction

1.1. Background and motivation

Along with the technological advances in digitalization and informatization, data centers (DTCs) have become one of the critical infrastructures, where their rapid deployments have led to a significant rise in power consumption [1,2]. In Europe, DTCs accounted for 2.7 % of the total electricity demand in 2018, which is predicted to be 3.2 % by 2030 [3]. In China, the total electricity consumption of DTCs is around 70 billion kWh, accounting for 1% of the national electricity consumption [4]. To address the above issue, we propose the coupled electricity-heat-computation system (CEHCS) as an efficient paradigm for planning DTCs, where renewable energy sources (RESs) are used for generating electricity, and waste heat recovery systems (WHRSs) are exploited to recycle the waste heat dissipated by servers, thereby improving energy efficiency. Generally, DTCs with CEHCSs represent green data centers (GDCs). From the perspective of GDC energy scheduling, power flow, heating flow, and computational workloads are collaboratively dispatched to consume RESs and recover waste heat as much as possible. As for the GDC planning, its core issue is to reasonably determine sites and sizes of RES investment facilities based on potential forecasts of load demands. In brief, CEHCS applications have become a research highlight for energy-saving in DTCs.

However, RESs used in GDCs demonstrate intense fluctuation and intermittence owing to the changes in weather conditions and geographic dispersions of RESs, e.g., wind turbines (WTs) and photovoltaic (PV) systems. Meanwhile, load demand, i.e., electricity, heating, and computational workload, also exhibits variations under the influence of economy, climate and lifestyle habits. In practice, source-load uncertainties are identified as one of the main technical challenges in long-term planning and short-term dispatch of GDCs.

1.2. Related work

1.2.1. Coupled Electricity-heat-computation

In recent years, computational workload flexibility has enabled GDCs to participate in electric power system scheduling. In [5], a two-stage scheduling model based on distributionally robust optimization was proposed for GDCs to reduce costs. An optimal planning model was proposed in [6] to determine DTC location and capacity in a distribution network, which also addressed the RES availability. The work in [7] developed a spatio-temporal scheduling model for the GDC workload with a focus on optimizing the RES utilization.

Meanwhile, DTC equipment produces much of the waste heat during operation, which also greatly affects energy efficiency and has the potential for recycling. In [8], DTCs were modeled as energy prosumers from the perspective of electricity consumption and heat production, where the heat production is represented by considering the complete dynamic heat flow, including the heat dissipation, exchange, recovery, and upgrading. A DTC-based energy hub model was developed in [9], with highly flexible energy conversion, consumption and storage. Especially, waste heat recovery was considered, which achieved better economic and environmental benefits. The authors in [10,11] constructed DTC integrated energy systems which served as prosumers and proposed cooperative optimization strategies for computing power, electricity and heat.

Although previous studies have explored CEHCS, they mainly focused on the coordinated scheduling of computational requests and multi-energy flows. A DTC without CEHCS, which is not equipped with RESs, BESSs, or WHRSs, exhibits low energy efficiency and requires infrastructure upgrades to improve its operation. Also, existing studies managed the indoor temperature of DTCs through the thermodynamic modeling of indoor air [8,9,11,12], while the server thermal management was not modeled in detail. Accordingly, our work focuses on the expansion planning of GDCs in which we consider the safe temperature constraints of CPUs to further ensure DTC reliability.

1.2.2. Characterization of source-load uncertainty

To deal with the source-load randomness, stochastic optimization is adopted, which requires multiple scenarios for characterizing system uncertainties [13]- [14]. The commonly adopted approaches for scenario generation include statistics-based, time series model-based, and data-driven methods. In general, for statistics-based methods, it is assumed that the data follows a certain probabilistic distribution, and then scenarios are obtained by sampling from the presumed probabilistic distribution, e.g., Monte Carlo sampling (MCS) [6,15–17], Latin hypercube sampling (LHS) [18] and Copula function-based method [19]. The statistics-based methods are intuitive and efficient. However, historical data must be assumed to follow a specific probabilistic distribution, while the statistical assumptions may not hold in practice.

Alternatively, time series model-based methods, e.g., state space model-based method [20], auto-regression and moving average model (ARMA) and autoregressive integrated moving average model (ARIMA) [21], are used to capture interdependent characteristics between time slots. The time series models can effectively depict RES temporal behaviors, with conciseness and interpretability. Nevertheless, time series models are only suitable for univariate time series analysis and cannot simultaneously consider the correlations between multiple variables.

In recent years, with the increasing prosperity of artificial intelligence-based techniques, data-driven methods have been widely exploited by adopting machine learning models, which demonstrate strong applicability [22]. Regardless of the data type and complexity, generative models can extract statistical features from historical observations and generate corresponding scenarios if there is sufficient data [23]. Currently, the generative adversarial network (GAN) model has become one of the most promising data-driven solutions for scenario generation. It spontaneously captures the internal characteristics of real scenarios by implicitly learning the data feature distribution, without relying on explicit probabilistic density function modeling. Many variants of the GAN model have been developed, such as deep convolutional GAN [24], Wasserstein-based GAN [25], controllable GAN [26] and style GAN [27]. In short, GAN-based models further improve the quality of generated scenarios and have become one of the most commonly adopted methods.

However, the training process for GANs is adversarial and unstable. Hence, previous studies exploited alternative models to provide stable performances. As a training-stable implicit generative model, the denoising diffusion probabilistic model (DDPM), also known as the diffusion model for simplicity, is considered to achieve high-quality samples in a wide variety of data modalities [28–30]. The diffusion model was initially used in the field of image synthesis, and some studies

have recently adopted it to the field of power systems. The studies in [31–33] exploited diffusion models for scenario generation of PV, wind power and extreme load scenarios, which captured the fluctuation characteristics and achieved excellent posterior representation. The posterior probabilistic distribution function of the diffusion model was redesigned in [34], which increased the diversity of the generated scenarios. In [35], a conditional diffusion-based scenario generation was proposed for controllable joint source-load scenario generation. The aforementioned studies have successfully exploited the potentials of diffusion models, which are straightforward to define and efficient to train. Meanwhile, diffusion models have been proven to have the ability to generate high-quality samples. Therefore, this work adopts diffusion models to characterize source-load uncertainties.

It is noted that diffusion models are commonly used in image processing, and their denoising networks are mostly formulated into a two-dimensional model. Nevertheless, CEHCS scenarios are one-dimensional time series, which means that the integrity of the time sequence will be destroyed if reshaped into a two-dimensional structure. In addition, CEHCS uncertainties come from a wide range of sources and involve multivariate time series, requiring joint scenario generation to ensure the rationality of multivariate scenarios. Thus, it is necessary to design a denoising network suitable for multivariate one-dimensional time series, which would be applicable to the proposed DTC model.

1.3. Technical contributions

To promote the infrastructure upgrade of DTCs, this work proposes a heuristic bi-level scheme for CEHCS expansion planning based on data-driven scenario generation. The comparison of this work with existing research is given in **Table 1**. The technical contributions of this work are summarized as follows:

1) A diffusion model based operational scenario generation (DMOSG) method is presented to generate multivariate scenarios, which incorporates a one-dimensional U-net (1dUN) to capture the inherent patterns from time series. The DMOSG method can characterize the multi-dimensional source-load uncertainties and enhance the cost-effectiveness and power usage efficiency of planning results.
2) A CEHCS expansion planning model is formulated to upgrade the DTC, aiming at optimizing its performance cost and energy efficiency. Especially, the safe temperature of server CPUs is considered during the modeling, and the operational status of each server is characterized individually, thus making the model higher-fidelity.
3) A heuristic bi-level iterative scheme is designed to cope with the nonlinearity of the constructed server operation model. The upper level adopts the particle swarm optimization (PSO) algorithm to determine the number of newly added servers, making the CEHCS model linear. The lower level directly solves the DTC planning model with CEHCS based on the server number obtained from the upper level.
4) Chance constraints of energy storage systems are proposed to cope with risks during CEHCS dispatch. In addition to the proposed DMOSG method applied for modeling the input scenarios for representing planning scenarios, energy dispatch uncertainties are considered via chance constraints to hedge against operational risks.

The remainder of this work is organized as follows. Section 2 introduces the DMOSG method in detail. Section 3 demonstrates the bi-level expansion planning approach. The case study and numerical results are given in Section 4. Finally, Section 5 provides the conclusive remarks.

2. Proposed diffusion model for operational scenario generation

Fig. 1 illustrates the proposed DMOSG method consisting of a forward process, training process and reverse process. The training sets can be obtained from the forward process, which is used to train the denoising network in stage II. The well-trained denoising network is applied in the reverse process to generate scenarios.

2.1. Forward process of DMOSG

The initial data ($\mathbf{x}_0$) is continuously added with Gaussian noise until it becomes pure noise during the forward process, i.e. diffusion process, which admits obtaining $\mathbf{x}_s$ at an arbitrary step s, as given in (1) [28].

$$\mathbf{x}_s = \sqrt{\overline{\alpha}_s}\mathbf{x}_0 + \sqrt{1-\overline{\alpha}_s}\boldsymbol{\varepsilon}_s \tag{1}$$

where $\alpha_s := 1-\beta_s$, $\overline{\alpha}_s := \prod_{i=1}^{s}\alpha_i$, $\boldsymbol{\varepsilon}_s \sim \mathcal{N}(\mathbf{0},\mathbf{I})$, s represents the diffusion step, $\mathbf{x}_0$ represents the data without adding noise and $\mathbf{x}_s$ denotes the data that has been added noise s times. It is noted that β_s denotes forward process variance, which is set to constants increasing linearly from β_1 to β_{M}. We use the notation M to denote the number of diffusion steps. The detailed derivation of the forward process is given in Appendix A.

2.2. Training process of DMOSG

The training process of the denoising network is given in **Algorithm 1** [28]. In the iterative process, step s is randomly selected from $\mathrm{Uniform}(\{1,2,\cdots,\mathrm{M}\})$ and specified through sinusoidal position embedding, then it is input into the denoising network together with $\mathbf{x}_s$. The predicted noise output by the model is compared with the real noise, thus updating the weights based on the loss function until convergence. It can be seen in **Fig. 1** that the U-net is chosen as the denoising network, which is a convolutional neural network named after its U-shaped structure and includes two processes: down-sampling and up-sampling. Due to its relatively low computational overhead, the U-net is frequently adopted in the diffusion model.

Algorithm 1. Denoising Network Training Processes.

Table 1
Comparison of this work with existing research

References	Focus	Research object	Individual modelling of servers	CPU temperature	Uncertainty handling	Model property	Nonlinearity handling
[5]	Operation of existing DTCs	Existing GDCs	/	/	MCS	Linear	/
[6]	Planning of newly built DTCs	Newly built DTCs	/	/	MCS	Linear	/
[7]	Operation of existing DTCs	Existing GDCs	/	/	Deterministic	Linear	/
[9]	Operation of existing DTCs	Existing GDCs	/	/	Deterministic	Linear	/
[10]	Operation of existing DTCs	Existing GDCs	/	/	Deterministic	Linear	/
[36]	Operation of existing DTCs	Existing GDCs	/	/	Deterministic	Nonlinear	Convex relaxation and piecewise linearization
this work	CEHCS upgrade of existing inefficient DTCs	Existing inefficient DTCs	Yes	Yes	DMOSG	Nonlinear	Heuristic bi-level solving scheme

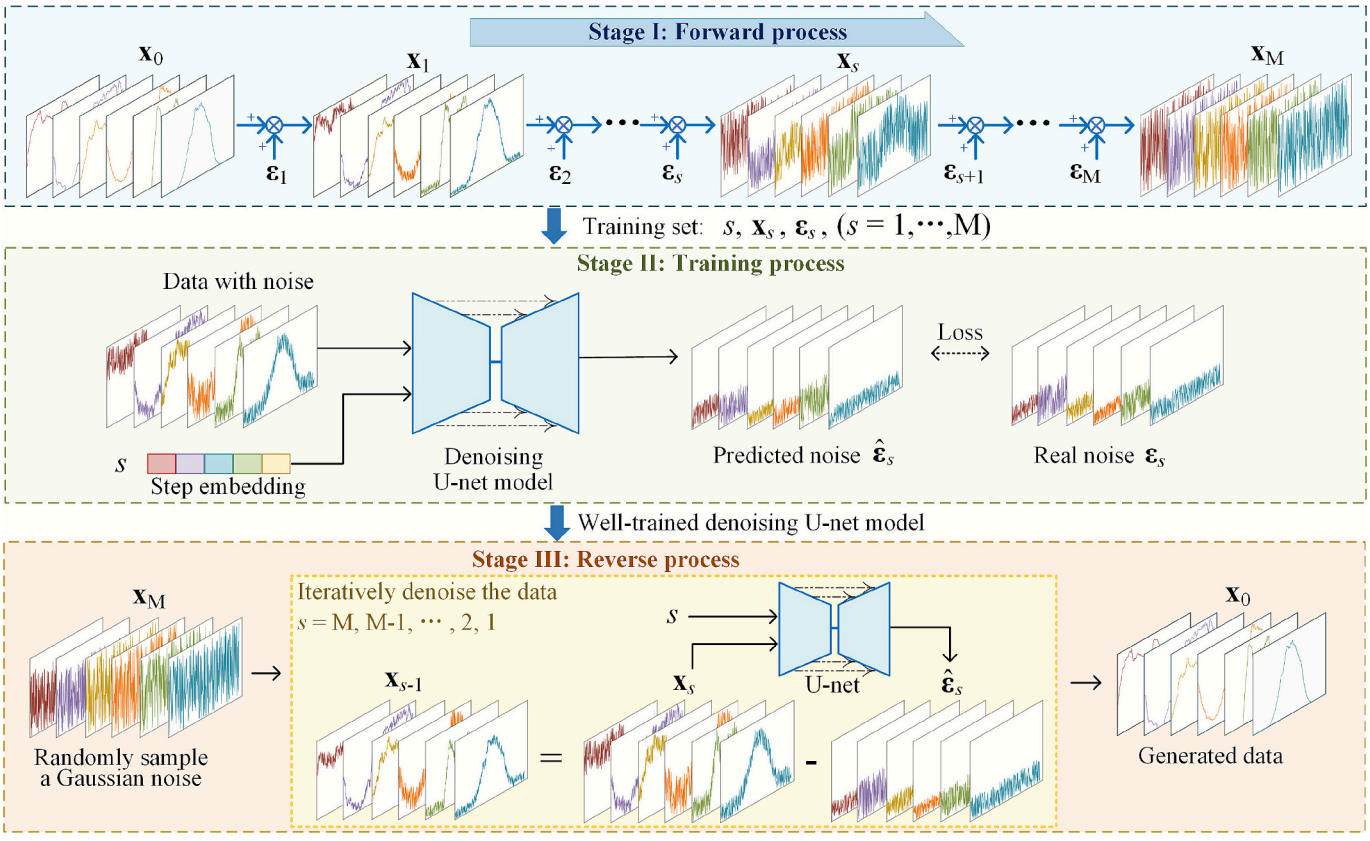


Fig. 1. Illustration of the diffusion model for the operational scenario generation method.

Algorithm 1: Denoising Network Training Processes.

Input: Batch size B, learning rate lr and number of training epochs N_{epoch}.

Initialize: Weight (θ) of the denoising network.
for $epoch = 1, 2, \cdots, N_{\text{epoch}}$ **do**
  Reset the gradients of the weights of the denoising network;
  Sample batch B scenarios from real data;
  Sample batch B steps $s \sim \text{Uniform}(\{1, 2, \cdots, M\})$;
  Sample a Gaussian noise $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$;
  # Update weight (θ) of denoising network
  $\theta \leftarrow \nabla_\theta \left\| \boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_\theta(\sqrt{\bar{\alpha}_s}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_s}\boldsymbol{\varepsilon}, s) \right\|^2$.
end

Fig. 2 illustrates the structure of the proposed 1dUN. To adapt to the multivariate time series in the CEHCS, the input format of the 1dUN is modified as one-dimensional sequences with multiple channels (6 channels in this work), where the length of each sequence corresponds to the length of the time series (the length of the time series is 24 in this work). During the down-sampling process, the one-dimensional convolution (Conv1d) is used to reduce the sequence length and increase the number of channels for extracting features. While in the up-sampling process, the one-dimensional transpose convolution is adopted to restore the initial length of time sequences. Concurrently, the features obtained from down-sampling are spliced into the features in the up-sampling process, thus achieving feature fusion of different levels and helping the 1dUN better learn the inherent characteristics of the training set.

Additionally, as a deep neural network, U-net faces a degradation problem due to the difficulty in implementing identity transformations. Excessive layers lead to a large number of activation functions, which map data to an inexplicable latent space and make it difficult to exploit initial features for neural networks. In other words, the deepening of the neural network results in too many nonlinear transformations, making it

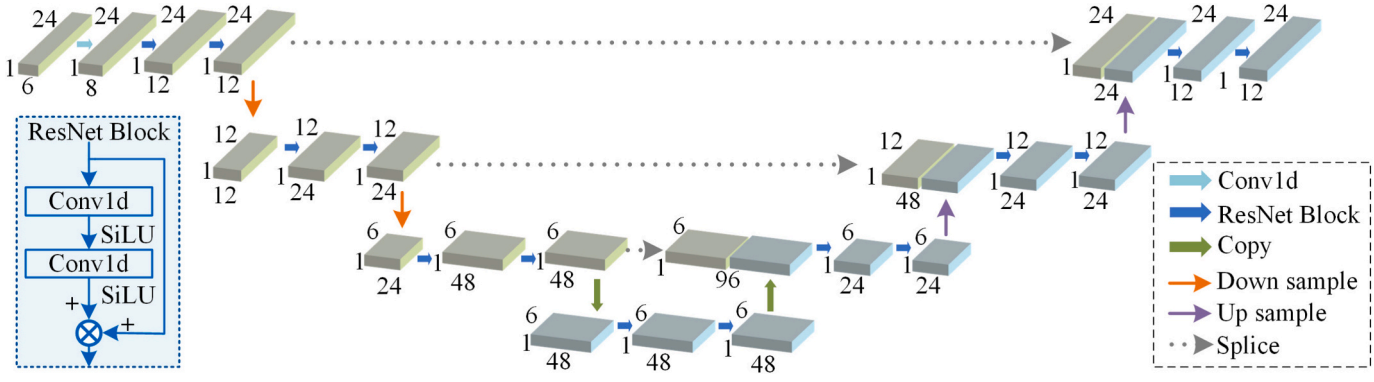


Fig. 2. Structure of the one-dimensional U-net.

hard to achieve a simple linear transformation. As such, this work incorporates the residual neural network (ResNet) block into the U-net, which contains identity mapping and helps the denoising network remember the initial characteristics of input data. It is considered that the ResNet can identify a balance between nonlinear and linear mapping, thereby improving denoising performance and preventing degradation.

Note that the proposed DMOSG method learns the data distribution implicitly rather than through explicit modeling. Thus, the representation of multi-dimensional source–load uncertainties is not directly visible. In practice, the 1dUN architecture captures correlations among different time series via its convolutional structure.

2.3. Reverse process of DMOSG

In the reverse process, Gaussian noise ($\mathbf{x}_M$) is denoised step by step to obtain the required scenario ($\mathbf{x}_0$), which is a reverse probability problem and needs to be solved using the Bayesian formula, as given in (2).

$$q(\mathbf{x}_{s-1}|\mathbf{x}_s,\mathbf{x}_0)=q(\mathbf{x}_s|\mathbf{x}_{s-1},\mathbf{x}_0)\frac{q(\mathbf{x}_{s-1}|\mathbf{x}_0)}{q(\mathbf{x}_s|\mathbf{x}_0)} \tag{2}$$

where $q(\mathbf{x}_s|\mathbf{x}_{s-1},\mathbf{x}_0)$, $q(\mathbf{x}_{s-1}|\mathbf{x}_0)$ and $q(\mathbf{x}_s|\mathbf{x}_0)$ are forward probability problems and can be easily obtained from the forward diffusion process. Thus, the posterior distribution $q(\mathbf{x}_{s-1}|\mathbf{x}_s,\mathbf{x}_0)$, where σ_s^2 and $\boldsymbol{\mu}_s$ are calculated by (3) and (4), respectively.

$$\sigma_s^2=\beta_s(1-\overline{\alpha}_{s-1})/(1-\overline{\alpha}_s) \tag{3}$$

$$\boldsymbol{\mu}_s=\frac{1}{\sqrt{\alpha_s}}\left(\mathbf{x}_s-\frac{1-\alpha_s}{\sqrt{1-\overline{\alpha}_s}}\boldsymbol{\varepsilon}_s\right) \tag{4}$$

Then the reverse process, i.e. generation process, can be implemented by **Algorithm 2** [28]. During each iteration, the well-trained model is adopted to predict the noise of step s, and then the forecasted noise ($\boldsymbol{\varepsilon}_\theta(\mathbf{x}_s,s)$) is used to calculate the mean value ($\boldsymbol{\mu}_s$) of the previous data, plus the standard deviation (σ_s) to obtain the data of the previous step ($\mathbf{x}_{s-1}$). The above process is continuously iterated until the required data ($\mathbf{x}_0$) is generated. The detailed derivation of the posterior distribution is given in Appendix B.

In the aforementioned discussion, it is clarified that the generation process of the diffusion model is achieved through continuous denoising, and the noise subtracted at each step is predicted through a denoising network. However, the inherent motive of the reverse process is not explained. Thus, some details are given to explain its significance.

Algorithm 2. Generation processes.

Algorithm 2: Generation processes.

Input: Number of generated scenarios N_{ge} and the well-trained denoising network.

Initialize: Randomly sample N_{ge} Gaussian noises $\mathbf{x}_M$.
for each $\mathbf{x}_M$ **do**
  for $s=$ M, M-1, ⋯, 2, 1 **do**
    Sample $\mathbf{z}\sim\mathcal{N}(\mathbf{0},\mathbf{I})$ **if** $s\geq 2$ **else** $\mathbf{z}=\mathbf{0}$;
    $\mathbf{x}_{s-1}=\frac{1}{\sqrt{\alpha_s}}\left(\mathbf{x}_s-\frac{1-\alpha_s}{\sqrt{1-\overline{\alpha}_s}}\boldsymbol{\varepsilon}_\theta(\mathbf{x}_s,s)\right)+\sigma_s\mathbf{z}$.
  end
end
return $\mathbf{x}_0$

From the forward process, it can be directly known that $\mathbf{x}_s$ follows the Gaussian distribution shown in (5).

$$\mathbf{x}_s\sim\mathcal{N}(\sqrt{\overline{\alpha}_s}\mathbf{x}_0,(1-\overline{\alpha}_s)\mathbf{I}) \tag{5}$$

Hence, the probabilistic distribution of $\mathbf{x}_s$ satisfies (6).

$$q\left(\mathbf{x}_s|\mathbf{x}_0\propto\exp\left(-\frac{1}{2}\left(\frac{(\mathbf{x}_s-\sqrt{\overline{\alpha}_s}\mathbf{x}_0)^2}{1-\overline{\alpha}_s}\right)\right)\right) \tag{6}$$

which can be calculated for the logarithmic gradient, as given in (7).

$$\nabla_{\mathbf{x}_s}\log q(\mathbf{x}_s|\mathbf{x}_0)=-\frac{\mathbf{x}_s-\sqrt{\overline{\alpha}_s}\mathbf{x}_0}{1-\overline{\alpha}_s} \tag{7}$$

where $\mathbf{x}_s$ can be substituted by (1), thereby deriving (8).

$$\nabla_{\mathbf{x}_s}\log q(\mathbf{x}_s|\mathbf{x}_0)=-\frac{1}{\sqrt{1-\overline{\alpha}_s}}\boldsymbol{\varepsilon}_s \tag{8}$$

It can be observed that the gradient $\mathbf{x}_s$ is proportional to the noise ($\boldsymbol{\varepsilon}_s$). In other words, the denoising network of diffusion model estimates the gradient of the data distribution by estimating noise, thus the following remarks can be drawn: a) the denoising process is equivalent to advancing $\mathbf{x}_s$ along the gradient direction of the probabilistic distribution, ultimately reaching the desired actual probabilistic distribution; b) the practical meaning of the output manifold space of the denoising network is the logarithmic gradient of the target data distribution.

3. Proposed bi-level expansion planning solution using scenario generation

3.1. Methodology overview

Fig. 3 illustrates the organization of the proposed solution, which consists of three parts, i.e. scenario generation and clustering, a heuristic algorithm based bi-level planning scheme and the planning result analysis. Especially, it should be noted that the bi-level planning is located in the middle part of Fig. 3, where the upper level determines the server capacity configuration and the lower level directly solves the CEHCS expansion planning model. The top part and bottom part of Fig. 3 respectively refer to the scenario generation and planning result analysis.

Scenario generation and clustering. First, a large number of scenarios are generated by the DMOSG method proposed in Section 2, thus fully characterizing the source-load uncertainties. However, since many scenarios lead to a long solution time, the generated scenarios are clustered through the k-medoids algorithm, thereby obtaining the representative scenarios. Then the representative scenarios are adopted as input parameters in the CEHCS expansion planning model.

Heuristic algorithm based bi-level planning scheme. Section 3.2 constructs the system model, and Section 3.3 formulates the DTC expansion planning model with CEHCS. It should be noted that the server model (10) and workload processing model (24) in Section 3.2 are nonlinear, because the number of servers ($Q^{srv,all}$) is a variable. Thus, the CEHCS expansion planning model cannot be directly solved by the Gurobi solver. To address this issue, this work develops a heuristic algorithm based bi-level planning scheme as follows.

In the upper level, the configured number of servers (Q^{srv}) is determined by the PSO algorithm and transferred to the lower level as a constant. Once Q^{srv} in (11) is fixed, $Q^{srv,all}$ becomes a constant, which makes (10) and (24) linear and enables the CEHCS expansion planning model to be directly solved by the Gurobi solver in the lower level.

In the lower level, the performance cost under the fixed server number (Q^{srv}) is obtained by solving the planning model and feeding back the solution to the upper level, according to which the PSO algorithm will adjust the number of configured servers, thus achieving better performance cost.

The above-mentioned interaction process between the two levels iteratively continues until convergence. The detailed iteration process is given in Section 3.4.

Note that the bi-level programming scheme introduced in this study differs from conventional robust optimization and two-stage stochastic optimization methods. Firstly, the proposed bi-level framework is

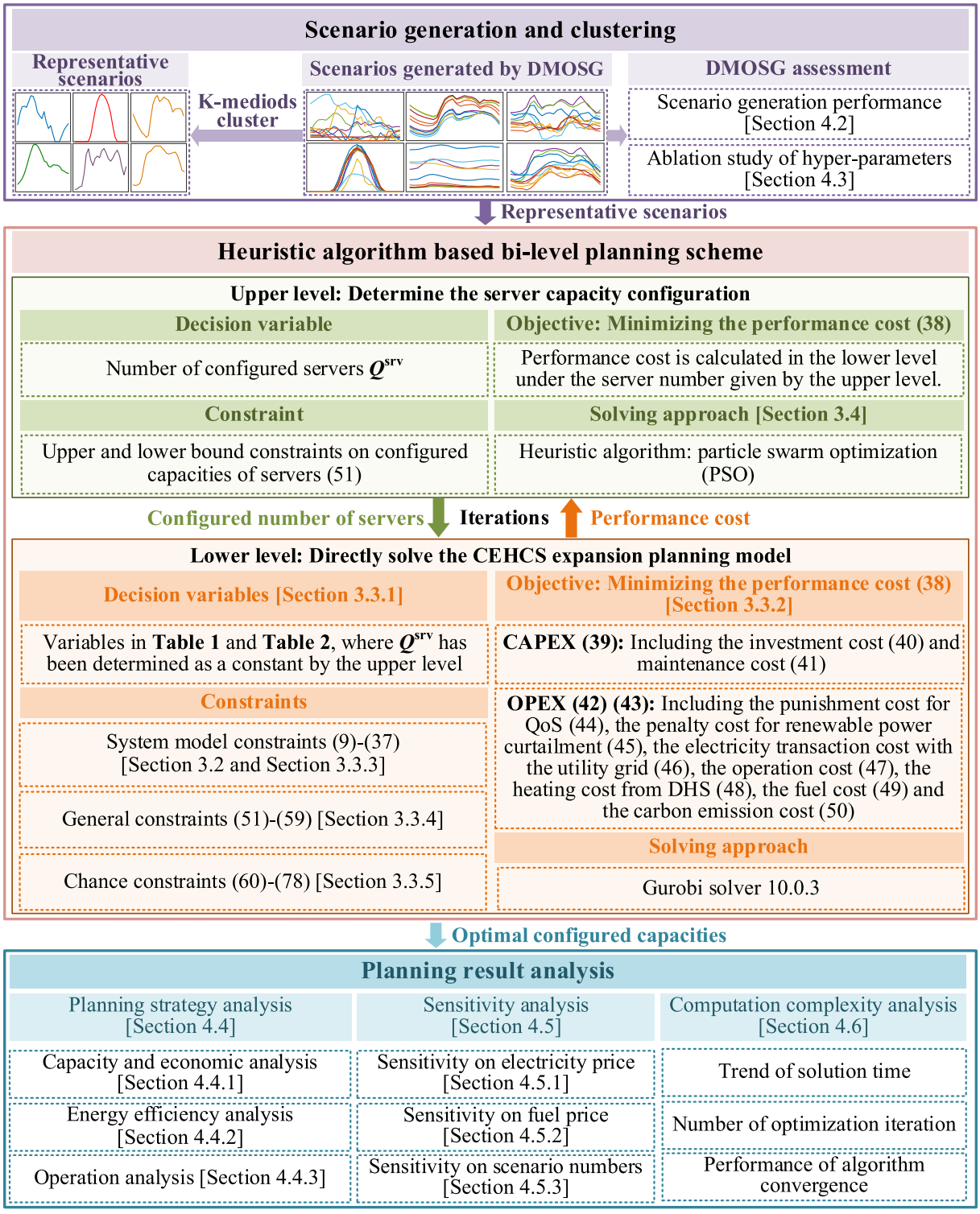


Fig. 3. Flowchart of the proposed solution.

specifically designed to address the nonlinearity of the server model, whereas traditional bi-level approaches are often employed for privacy protection. The conventional approaches are unable to process the non-linearity of the server model established in this work. Further, the upper level is not a strict optimization process, but a process of fixing the number of servers according to PSO's search strategy. As for outcomes, the proposed bi-level scheme is not as conservative as robust optimization, while also considering uncertainty more fully than traditional two-stage stochastic optimization, thus exhibiting better performance cost than the other two kinds of approaches.

Analysis of planning results. As shown in the bottom part of Fig. 3, the planning results are validated through a comprehensive analysis, involving planning strategy analysis (Section 4.4), sensitivity analysis (Section 4.5) and computation complexity analysis (Section 4.6).

3.2. System model

Fig. 4 demonstrates the CEHCS model, which is connected with the

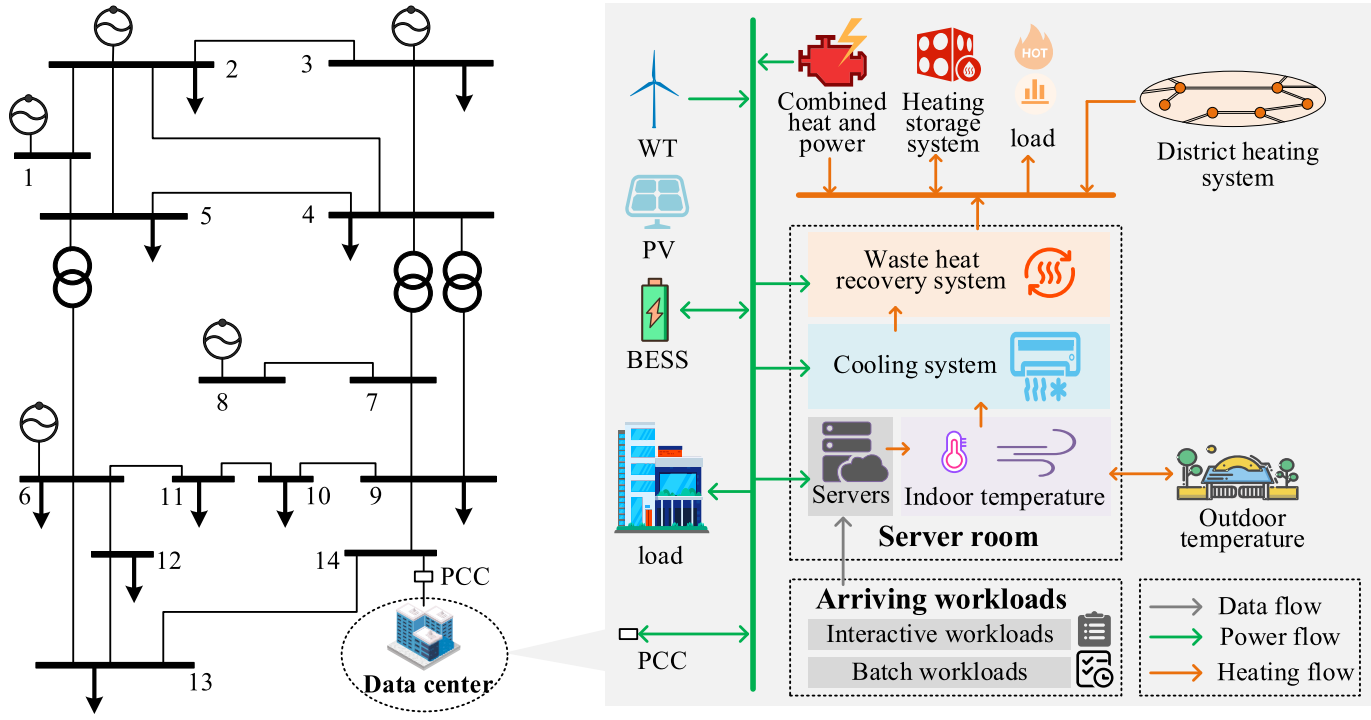


Fig. 4. Coupled Electricity-heat-computation system.

distribution network through a point of common coupling (PCC). Before the infrastructure upgrade, the electric and heating loads of the DTC are supplied by the utility grid and district heating system, respectively, with no other energy sources. Meanwhile, the equipment in the DTC only includes servers and cooling systems. However, in the planned electricity-heat-computation coupling system, various types of facilities that contribute to energy saving are considered, i.e. PVs, WTs, BESSs, WHRSs, CHPs and HSSs. Note that the planning scheme also includes the expansion of servers and cooling systems to cope with the future data traffic increase. In this work, the recovered waste heat is utilized locally, and hence, the topology of DHS is not discussed here.

3.2.1. *Energy consumption model of data center*

The electricity demand ($P_{i,t}^{\mathrm{DC}}$) in GDC represents IT equipment ($P_{i,t}^{\mathrm{IT}}$), cooling systems ($P_{i,t}^{\mathrm{cool}}$) and WHRSs ($P_{i,t}^{\mathrm{whrs}}$), while other electrical loads ($P_{i,t}^{\mathrm{other}}$) are also considered, as shown in (9).

$$P_{i,t}^{\mathrm{DC}} = P_{i,t}^{\mathrm{IT}} + P_{i,t}^{\mathrm{cool}} + P_{i,t}^{\mathrm{whrs}} + P_{i,t}^{\mathrm{other}} \tag{9}$$

where i and t represent the index of representative days and time slots, respectively. Further, $P_{i,t}^{\mathrm{IT}}$ and $P_{i,t}^{\mathrm{cool}}$ can be calculated by (10) and (16), respectively.

$$P_{i,t}^{\mathrm{IT}} = \sum\nolimits_{j=1}^{Q^{\mathrm{srv,all}}} P_{ij,t}^{\mathrm{srv}} \tag{10}$$

$$Q^{\mathrm{srv,all}} = Q^{\mathrm{srv,exist}} + \qquad Q^{\mathrm{srv}} \tag{11}$$

$$P_{ij,t}^{\mathrm{srv}} = \wp_{ij,t} \cdot \left[P^{\mathrm{idle}} + \left(P^{\mathrm{peak}} - P^{\mathrm{idle}} \right) \cdot \mu_{ij,t} \right] \tag{12}$$

$$\wp_{ij,t} = \begin{cases} 1, \text{if } \ell_{ij,t}^{\mathrm{srv}} = 0 \\ 0, \text{else} \end{cases} \tag{13}$$

$$\ell_{ij,t}^{\mathrm{srv}} \geq 0, \forall i,j,t \tag{14}$$

$$\mu_{ij,t} = \zeta^{\mathrm{req}} \cdot \ell_{ij,t}^{\mathrm{srv}} \tag{15}$$

where j is the index of servers, $Q^{\mathrm{srv,all}}$ represents the number of servers, $Q^{\mathrm{srv,exist}}$ denotes the number of existing servers, and Q^{srv} represents the number of new servers to be configured. $P_{ij,t}^{\mathrm{srv}}$ represents the power consumed by a server, P^{idle} represents the idle power of a server, P^{peak} represents the peak power of a server, $\mu_{ij,t}$ represents the utilization rate of servers. Note that $\wp_{ij,t}$ is a Boolean decision variable, and $\wp_{ij,t} = 1$ indicates that the server is in an active state. Otherwise, it indicates a shutdown state. $\ell_{ij,t}^{\mathrm{srv}}$ denotes the number of requests being processed by the server, and ζ^{req} denotes the CPU utilization required for a single computation request.

$$P_{i,t}^{\mathrm{cool}} = \beta_1 H_{i,t}^{\mathrm{cool}} + \beta_2 \tag{16}$$

where $H_{i,t}^{\mathrm{cool}}$ denotes the heat extracted by the cooling system, β_1 and β_2 are empirical coefficients. The operational constraint of cooling systems is given in (17) [8].

$$\beta_2 \leq P_{i,t}^{\mathrm{cool}} \leq Q^{\mathrm{cool,exist}} + Q^{\mathrm{cool}} \tag{17}$$

where $Q^{\mathrm{cool,exist}}$ and Q^{cool} denotes the existing capacity and configured capacity of cooling systems, respectively.

3.2.2. *Computational workload model*

The computational workloads that arrive at data centers can be split into two types: a) interactive workloads and b) batch workloads. Interactive workloads are delay-sensitive and required to be processed immediately upon arrival, while batch workloads are delay-tolerant and can be delayed for execution [37]. Interactive workloads are modeled as (18).

$$\lambda_{i,t}^{\mathrm{inter}} \geq 0 \tag{18}$$

where $\lambda_{i,t}^{\mathrm{inter}}$ denotes the arrival rate. Nevertheless, batch workloads are formulated as (19)-(23) [11].

$$\lambda_{i,\tau,t}^{\mathrm{batch}} \geq 0, \tau \leq t \tag{19}$$

$$\sum_{t=1}^{T} \lambda_{i,\tau,t}^{\text{batch}} = \lambda_{i,\tau}^{\text{batch,total}} \tag{20}$$

$$\sum_{t=\tau}^{t_{\tau}^{\text{ddl}}} \lambda_{i,\tau,t}^{\text{batch}} = \lambda_{i,\tau}^{\text{batch,total}} \tag{21}$$

$$t_{\tau}^{\text{ddl}} = \min\left(\tau + t_{\tau}^{\text{delay,max}}, T\right) \tag{22}$$

$$\lambda_{i,t}^{\text{batch}} = \sum_{\tau=1}^{t} \lambda_{i,\tau,t}^{\text{batch}} \tag{23}$$

where $\lambda_{i,\tau,t}^{\text{batch}}$ denotes the number of batch workloads that arrive at time τ and then are processed at time t, $\lambda_{i,\tau}^{\text{batch,total}}$ denotes the total number of batch workloads that arrive at time τ, t_{τ}^{ddl} denotes the deadline of batch workloads that arrived at time τ, $t_{\tau}^{\text{delay,max}}$ denotes the maximum allowed delay time for the batch workloads that arrive at time τ, $\lambda_{i,t}^{\text{batch}}$ denotes the total number of batch workloads being processed at time t. (20) and (21) ensure that batch workloads are completed before the deadline, thereby ensuring the quality of service (QoS).

The number of workloads processed by the DTC is equal to the sum of the workloads processed on each server, as shown in (24).

$$\lambda_{i,t}^{\text{inter}} + \lambda_{i,t}^{\text{batch}} = \sum\nolimits_{j=1}^{Q^{\text{srv,all}}} \ell_{i,j,t}^{\text{srv}} \tag{24}$$

The queuing delay of computational workloads is required to meet the following constraint based on the M/M/1 queuing theory, as given in (25).

$$\frac{1}{N_{i,t}^{\text{srv}} \lambda^{\text{srv,max}} - \lambda_{i,t}^{\text{inter}} - \lambda_{i,t}^{\text{batch}}} + t^{\text{trans}} \le t^{\text{inter,max}} \tag{25}$$

where $N_{i,t}^{\text{srv}}$ denotes the number of active servers, $\lambda^{\text{srv,max}}$ denotes the maximum computing rate of servers, t^{trans} denotes the transmission delay, and $t^{\text{inter,max}}$ denotes the maximum allowed delay time for interactive workloads.

Furthermore, the number of active servers should follow the constraint (26).

$$0 \le N_{i,t}^{\text{srv}} \le Q^{\text{srv,exist}} + Q^{\text{srv}} \tag{26}$$

where $Q^{\text{srv,exist}}$ denotes the number of existing servers in the DTC.

3.2.3. Ambient thermodynamic model

To fully explore the impact of environmental temperature on server operation, the impact of indoor and outdoor temperatures on the operation of DTCs is considered in this article and the thermal inertia of DTCs is modeled in detail. The heat transfer involved in this work includes three parts: (a) heat transfer between servers and indoor air, (b) heat transfer between cooling systems and indoor air, and (c) heat transfer between indoor and outdoor air. The relationship between heat transfer and temperature change can be modeled as a first-order linear difference equation, as given in (27) [9].

$$H_{i,t}^{\text{IT}} - H_{i,t}^{\text{cool}} + \frac{T_{i,t}^{\text{out}} - T_{i,t}^{\text{in}}}{R} = C^{\text{DC}} r_{i,t}^{\text{in}} \tag{27}$$

where $H_{i,t}^{\text{IT}}$ denotes the heat dissipation power of IT equipment, $T_{i,t}^{\text{in}}$ and $T_{i,t}^{\text{out}}$ denote the temperature of indoor air and outdoor air, respectively, R denotes the equivalent heat resistance, C^{DC} denotes the heat capacity and $r_{i,t}^{\text{in}}$ denotes the changing rate of indoor temperature. Almost all of the electricity consumed by servers is converted into waste heat, as shown in (28).

$$H_{i,t}^{\text{IT}} = \theta^{\text{IT}} P_{i,t}^{\text{IT}} \tag{28}$$

where θ^{IT} represents the heat dissipation rate of IT equipment and is set to 1.

The indoor temperature of DTCs must be controlled within a reasonable range to ensure the normal operation of IT equipment such as servers, as excessive temperature changes can lead to a decrease in the lifespan of IT equipment. Thus, the indoor temperature should be limited by (29) - (32).

$$r_{i,t}^{\text{in}} = T_{i,t}^{\text{in}} - T_{i,t-1}^{\text{in}}, \forall t \ge 2 \tag{29}$$

$$T_{i,1}^{\text{in}} = T^{\text{in,min}} \tag{30}$$

$$T^{\text{in,min}} \le T_{i,t}^{\text{in}} \le T^{\text{in,max}} \tag{31}$$

$$r^{\text{in,min}} \le r_{i,t}^{\text{in}} \le r^{\text{in,max}} \tag{32}$$

where $T_{i,1}^{\text{in}}$ represents the initial indoor temperature, $T^{\text{in,min}}$ and $T^{\text{in,max}}$ represents the minimum and maximum indoor temperature of DTCs, respectively, $r^{\text{in,min}}$ and $r^{\text{in,max}}$ represents the minimum and maximum changing rate of indoor temperature, respectively.

3.2.4. Server thermal management model

To further discuss the temperature requirements and constraints of the IT equipment, the allowable operating temperature of server CPUs is considered. The CPU dissipates much heat, which affects the inside temperature of the server. The CPU temperature is impacted by not only its power consumption but also thermal resistance and heat capacity, along with the indoor temperature [38]. Thus, the CPU temperature ($T_{i,j,t}^{\text{cpu}}$) can be formulated by a resistance-capacitance model [39], as given in (33).

$$T_{i,j,t}^{\text{cpu}} = P_{i,j,t}^{\text{srv}} R^{\text{cpu}} + T_{i,t}^{\text{in}} + \left(T^{\text{cpu,ini}} - P_{i,j,t}^{\text{srv}} R^{\text{cpu}} - T_{i,t}^{\text{in}}\right) \mathrm{e}^{-t/(R^{\text{cpu}} C^{\text{cpu}})} \tag{33}$$

where R^{cpu} and C^{cpu} denote the thermal resistance and heat capacity of the CPU, respectively; $T^{\text{cpu,ini}}$ denotes the initial temperature of the CPU; e is a mathematical constant.

To ensure the operational reliability of servers, the CPU temperature cannot exceed the maximum allowable temperature ($T^{\text{cpu,max}}$), as shown in (34).

$$T_{i,j,t}^{\text{cpu}} \le T^{\text{cpu,max}} \tag{34}$$

3.2.5. Waste heat recovery model

The waste heat in data centers can be generally recovered through two methods: a hot aisle layout locally and a cooling system [29,30]. Considering that the heat captured from the hot aisle is dispersed and can hardly be utilized centrally in practice, this work recovers waste heat from the condenser of cooling systems. The waste heat recovery process is modeled as (35)-(37).

$$H_{i,t}^{\text{re}} = \theta^{\text{whrs}} P_{i,t}^{\text{whrs}} \tag{35}$$

$$0 \le P_{i,t}^{\text{whrs}} \le Q^{\text{whrs}} \tag{36}$$

$$0 \le H_{i,t}^{\text{re}} \le \omega^{\text{re}} H_{i,t}^{\text{cool}} \tag{37}$$

where $H_{i,t}^{\text{re}}$ represents the recovered heating power, θ^{whrs} represents the electricity to heat ratio of WHRSs, Q^{whrs} represents the configured capacity of WHRSs, ω^{re} represents the waste heat recovery efficiency. Eq. (35) characterizes the efficiency of the WHRS in utilizing electricity, Eq. (36) shows that the WHRS needs to be operated within its configured capacity, and Eq. (37) means that the recovered heat is limited by cooling systems.

3.3. Problem formulation of DTC expansion planning with CEHCS

3.3.1. Decision variables

Table 2 shows the planning decision variables, i.e. the configured

Table 2
Planning decision variables

Equipment	Decision variables
Server	Q^{srv}
Cooling system	Q^{cool}
Waste heat recovery system	Q^{whrs}
Heating storage system	Q^{hss}
Combined heat and power	Q^{chp}
Battery energy storage system	Q^{es}
Wind turbine	Q^{wt}
Photovoltaic system	Q^{pv}

capacities of facilities to be installed in DTC. The scheduling decision variables are the operational status of various facilities and the processing rate of computational workloads, as given in **Table 3**.

3.3.2. Objective

The decision-making objective is to minimize the performance cost (C), including the CAPEX (F^{CAP}) and OPEX (F^{OP}), given in (38). The CAPEX involves the investment cost (C^{inv}_{equip}) and maintenance cost (C^{main}_{equip}), as defined in (39).

$$\min \quad C = F^{CAP} + F^{OP} \tag{38}$$

$$F^{CAP} = \sum_{equip}\left(C^{inv}_{equip} + C^{main}_{equip}\right) \tag{39}$$

where F denotes the scheduling costs; *equip* denotes the equipment type (servers, cooling systems, WHRSs, HSS, CHP, BESS, WT, and PV). The investment and maintenance cost of certain types of facilities is calculated by (40) and (41) respectively.

$$C^{inv}_{equip} = Q_{equip} \cdot \chi^{inv}_{equip} \cdot \frac{d(1+d)^{life}}{(1+d)^{life}-1} \tag{40}$$

$$C^{main}_{equip} = Q_{equip} \cdot m^{main}_{equip} \cdot life \tag{41}$$

where Q_{equip} denotes the configured capacity, χ^{inv}_{equip} denotes the unit investment price, d and *life* denote the discount rate and the life cycle of the facilities, respectively, m^{main}_{equip} denotes the unit maintenance cost.

The OPEX (F^{OP}) denotes the scheduling costs of all representative days, as shown in (42).

$$F^{OP} = 365 \cdot life \cdot \sum_{i}(p_i \cdot F_i) \tag{42}$$

where p_i and F_i denote the occurring probability and scheduling costs of the ith representative day. Specifically, F_i is calculated by (43).

$$F_i = F^{qos}_i + F^{penalty}_i + F^{g}_i + F^{o}_i + F^{dhs}_i + F^{fuel}_i + F^{carbon}_i \tag{43}$$

Table 3
Scheduling decision variables

Equipment	Decision variables
Server	$N^{srv}_{i,t}$
Cooling system	$P^{cool}_{i,t}$
Waste heat recovery system	$P^{whrs}_{i,t}$
Heating storage system	$H^{hs,c}_{i,t}$, $H^{hs,d}_{i,t}$
Combined heat and power	$P^{chp}_{i,t}$
Battery energy storage system	$P^{es,c}_{i,t}$, $P^{es,d}_{i,t}$
Wind turbine	$P^{wt}_{i,t}$
Photovoltaic system	$P^{pv}_{i,t}$
Utility grid	$P^{buy}_{i,t}$, $P^{sell}_{i,t}$
District heating system	$H^{dhs}_{i,t}$
Interactive workloads	$\lambda^{inter}_{i,t}$
Batch workloads	$\lambda^{batch}_{i,r,t}$

where F^{qos}_i denotes the punishment cost for QoS not met, $F^{penalty}_i$ denotes the penalty cost for wind power curtailment and PV power curtailment, F^{g}_i denotes the electricity transaction cost with the utility grid, F^{o}_i denotes the operation cost, F^{dhs}_i denotes the heating cost from DHS, F^{fuel}_i denotes fuel cost and F^{carbon}_i denotes carbon emission cost. The above seven items of costs are calculated by (44) - (50), respectively.

$$F^{qos}_i = c^{qos}_{inter} N^{fail,inter}_i + c^{qos}_{batch} N^{fail,batch}_i \tag{44}$$

$$F^{penalty}_i = \sum_{t=1}^{T}\left(c^{penalty}_{pv} P^{pv,over}_{i,t} + c^{penalty}_{wt} P^{wt,over}_{i,t}\right) \tag{45}$$

$$F^{g}_i = \sum_{t=1}^{T}\left(c^{b}_{i,t} P^{g,b}_{i,t}\right) \tag{46}$$

$$F^{o}_i = \sum_{t=1}^{T}\sum_{equip} c^{o}_{equip} P^{equip}_{i,t} \tag{47}$$

$$F^{dhs}_i = \sum_{t=1}^{T}\left(c^{dhs} H^{dhs}_{i,t}\right) \tag{48}$$

$$F^{fuel}_i = \sum_{t=1}^{T}\left(c^{f,chp} P^{chp}_{i,t}\right) \tag{49}$$

$$F^{carbon}_i = \sum_{t=1}^{T}\left(c^{g,b,c}\rho^{g} P^{g,b}_{i,t} + c^{dhs,c}\rho^{dhs} H^{dhs}_{i,t} + c^{chp,c}\rho^{chp} P^{chp}_{i,t}\right) \tag{50}$$

In (44), c^{qos}_{inter} and c^{qos}_{batch} denote the unit punishment cost for QoS not met of interactive workloads and batch workloads, respectively, $N^{fail,inter}_i$ and $N^{fail,batch}_i$ denote the number of failed requests of interactive workloads and batch workloads, respectively. In (45), $c^{penalty}_{pv}$ and $c^{penalty}_{wt}$ denote the unit penalty cost of PV power curtailment and wind power curtailment, respectively, $P^{pv,over}_{i,t}$ and $P^{wt,over}_{i,t}$ denote the power of RES overproduction. In (46), $c^{b}_{i,t}$ and $c^{s}_{i,t}$ denote the purchasing and sales price of electricity, respectively, $P^{g,b}_{i,t}$ denotes the purchasing power of electricity. In (47), c^{o}_{equip} and $P^{equip}_{i,t}$ denote the unit operation cost and operational power of specified equipment (*equip* = PV, WT, BESS, HSS, WHRS, CHP, cooling systems and servers). In (48), c^{dhs} and $H^{dhs}_{i,t}$ denote the unit price and heating power of energy purchased from DHS, respectively. In (49), $c^{f,chp}$ and $P^{chp}_{i,t}$ denote the unit fuel price and electric power of CHP, respectively. In (50), $c^{g,b,c}$, $c^{dhs,c}$ and $c^{chp,c}$ denote the unit carbon emission cost of the utility grid, DHS, and CHP, respectively, and ρ^{g}, ρ^{dhs} and ρ^{chp} denote the emission factors.

3.3.3. System model constraints

Constraints on DTC energy consumption are given in (9)-(15), constraints on cooling systems are given in (16) and (17), constraints on computational workloads are given in (18)-(26), the ambient thermodynamical constraints are given in (27)-(32), constraints on CPU temperature are given in (33)-(34), and the constraints on WHRSs are given in (35)-(37).

3.3.4. General constraints

(1) Constraints on equipment capacities

The constraints on the upper and lower bounds of equipment capacities are considered in the optimal equipment capacity model.

$$0 \le Q_{equip} \le Q^{max}_{equip} \tag{51}$$

where Q^{max}_{equip} denotes the maximum configured capacity (*equip* = servers, cooling systems, WHRSs, HSS, CHP, BESS, WT, and PV).

(2) Constraints on direct-current power flow of the electric power system

In the IEEE standard test systems, the power balance of each bus

should be met, as shown in (52).

$$\sum_{s\in\Omega_n^{\text{source}}} P_{i,t,n,s} + \sum_{l\in\Omega_n^{\text{branch}}} P_{i,t,n,l}\cdot u_{n,l} = P_{i,t,n}^{\text{load}},\ \ n=1,2,\cdots,N_{\text{bus}} \tag{52}$$

where n is the index of buses, *source* denotes the power supply equipment, Ω_n^{source} denotes the set of sources connected to the bus n, $P_{i,t,n,s}$ denotes the electric power of the source, l denotes the branches of the electric power system, Ω_n^{branch} denotes the set of branches connected to the bus n, $P_{i,t,n,l}$ denotes the power flow of the branch l, $u_{n,l}$ denotes the direction of the power flow ($u_{n,l}=1$ when the current of branch l starts from the bus n, and $u_{n,l}=-1$ when the current of branch l flows into the bus n), $P_{i,t,n}^{\text{load}}$ denotes the electricity demand of bus n, and N_{bus} denotes the number of buses.

In the direct-current power flow model, the branch power is linearly correlated with the phase angles of connected buses, as shown in (53).

$$P_{i,t,j,k} = \frac{\theta_j - \theta_k}{x_{j,k}} \tag{53}$$

where $P_{i,t,j,k}$ denotes the branch power flowing from bus j to bus k, θ_j and θ_k denote the phase angles of bus j and bus k respectively, and $x_{j,k}$ denotes the reactance of the branch.

Generally, the phase angles of buses need to be controlled within a reasonable range to ensure the safety and stability of the power system, as shown in (54).

$$\theta_n^{\min} \le \theta_n \le \theta_n^{\max} \tag{54}$$

where θ_n denotes the phase angle of the bus n, $\theta_n^{\min}$ and $\theta_n^{\max}$ denote the minimum and maximum phase angles, respectively.

(3) Constraints on renewable generators

The output power of PVs and WTs is nonnegative and limited below the upper bound, as shown in (55) and (56).

$$0 \le P_{i,t}^{\text{pv}} \le \min\left\{P_{i,t}^{\text{pv,max}}, Q_{\text{pv}}\right\} \tag{55}$$

$$0 \le P_{i,t}^{\text{wt}} \le \min\left\{P_{i,t}^{\text{wt,max}}, Q_{\text{wt}}\right\} \tag{56}$$

where $P_{i,t}^{\text{pv,max}}$ and $P_{i,t}^{\text{wt,max}}$ represent the predicted power generation of PVs and WTs, respectively.

(4) Constraints on CHPs

For CHPs, the constraints on upper bound, ramping rate and electric-heating ratio are given in (57), (58), and (59), respectively. The constraints of on/off operational status can be found in [11], which are not shown here for the sake of space limitation.

$$0 \le P_{i,t}^{\text{chp}} \le Q^{\text{chp}} \tag{57}$$

$$-\varpi \le P_{i,t}^{\text{chp}} - P_{i,t-1}^{\text{chp}} \le \varpi \tag{58}$$

$$H_{i,t}^{\text{chp}} = \omega^{\text{chp}} P_{i,t}^{\text{chp}} \tag{59}$$

where ϖ represents the ramping rate, and ω^{chp} represents the electric-heating ratio of CHPs.

3.3.5. Chance constraints

(1) Chance constraints on BESSs

The operation of BESSs follows the constraints on the state of charge, the constraints on charging and discharging states, and the constraints on the electric power, as given in (60) - (62), (63), and (64) - (65), respectively.

$$Soc^{\min} \le Soc_{i,t} \le Q^{\text{es}} \tag{60}$$

$$Soc^{\min} = 0.1\cdot Q^{\text{es}} \tag{61}$$

$$Soc_{i,t} = Soc_{i,t\text{-}1}\cdot(1-\lambda^{\text{es}}) + \vartheta^{\text{es,c}}\cdot P_{i,t\text{-}1}^{\text{es,c}} - \frac{1}{\vartheta^{\text{es,d}}}\cdot P_{i,t\text{-}1}^{\text{es,d}} \tag{62}$$

$$\eta_{i,t}^{\text{es,c}} + \eta_{i,t}^{\text{es,d}} \le 1 \tag{63}$$

$$0 \le P_{\text{dis},i,t}^{\text{es}} \le P_t^{\text{es,d,max}}\cdot\eta_{i,t}^{\text{es,d}} \tag{64}$$

$$0 \le P_{i,t}^{\text{es,c}} \le P_t^{\text{es,c,max}}\cdot\eta_{i,t}^{\text{es,c}} \tag{65}$$

where $Soc^{\min}$ denotes the minimum state of charge, $Soc_{i,t}$ denotes the state of charge, λ^{es} denotes the self-loss rate, $\vartheta^{\text{es,c}}$ and $\vartheta^{\text{es,d}}$ represent the charge efficiency and discharge efficiency respectively, $\eta_{i,t}^{\text{es,c}}$ and $\eta_{i,t}^{\text{es,d}}$ denote the charging and discharging state respectively, $P_t^{\text{es,c,max}}$ and $P_t^{\text{es,d,max}}$ denote the maximum charging and discharging power respectively.

The aforementioned DMOSG is used for modeling the uncertainties of input scenarios, but the uncertainties during energy dispatch also deserve attention to hedge risks. Therefore, to better optimize the objective, chance constraints are adopted to flexibly adjust the maximum charging and discharging power, as shown in (66) and (67).

$$\Pr\left\{P_t^{\text{es,d,max}} = 0.7\cdot Q^{\text{es}}\right\} \ge \varepsilon^{\text{es}} \tag{66}$$

$$\Pr\left\{P_t^{\text{es,c,max}} = 0.7\cdot Q^{\text{es}}\right\} \ge \varepsilon^{\text{es}} \tag{67}$$

where $\Pr\{\}$ denotes the probability of the constraint being valid and ε^{es} is the confidence level.

(2) Chance constraints on HSSs

The operation of HSSs follows the constraints on the heat storage state, the constraint on heat absorption and release states, and the constraints on the heating power, as given in (68) -(69), (70), and (71) -(72), respectively.

$$Soh^{\min} \le Soh_{i,t} \le Q^{\text{hs}} \tag{68}$$

$$Soh_{i,t} = Soh_{i,t\text{-}1}\cdot\left(1-\lambda^{\text{hs}}\right) + \vartheta^{\text{hs,c}}\cdot H_{i,t\text{-}1}^{\text{hs,c}} - \frac{1}{\vartheta^{\text{hs,d}}}\cdot H_{i,t\text{-}1}^{\text{hs,d}} \tag{69}$$

$$\eta_{i,t}^{\text{hs,c}} + \eta_{i,t}^{\text{hs,d}} \le 1 \tag{70}$$

$$0 \le H_{i,t}^{\text{hs,d}} \le H_t^{\text{hs,d,max}}\cdot\eta_{i,t}^{\text{hs,d}} \tag{71}$$

$$0 \le H_{i,t}^{\text{hs,c}} \le H_t^{\text{hs,c,max}}\cdot\eta_{i,t}^{\text{hs,c}} \tag{72}$$

where $Soh^{\min}$ denotes the minimum heat storage state, $Soh_{i,t}$ denotes the heat storage state, λ^{hs} denotes the self-loss rate, $\vartheta^{\text{hs,c}}$ and $\vartheta^{\text{hs,d}}$ represent the heat absorption and heat release efficiency respectively, $\eta_{i,t}^{\text{hs,c}}$ and $\eta_{i,t}^{\text{hs,d}}$ denote the heat absorption and release state respectively, $H_t^{\text{hs,c,max}}$ and $H_t^{\text{hs,d,max}}$ denote the maximum heat absorption and release power respectively.

To flexibly supply heating load, the maximum heat absorption and release power are also formulated as chance constraints, as given in (73) and (74).

$$\Pr\left\{H_t^{\text{hs,d,max}} = 0.7\cdot Q^{\text{hs}}\right\} \ge \varepsilon^{\text{hs}} \tag{73}$$

$$\Pr\left\{H_t^{\text{hs,c,max}} = 0.7\cdot Q^{\text{hs}}\right\} \ge \varepsilon^{\text{hs}} \tag{74}$$

(3) Deterministic transformation of chance constraints

Boolean decision variables (γ_t^{es},γ_t^{hs}) are introduced to reformulate the abovementioned chance constraints (66), (67), (73) and (74) into deterministic constraints, as shown in (75) -(78).

$$P_t^{\text{es,c,max}} = P_t^{\text{es,d,max}} = 0.7\cdot Q^{\text{es}} + \left(1-\gamma_t^{\text{es}}\right)\cdot\left(0.9\cdot Q^{\text{es}} - 0.7\cdot Q^{\text{es}}\right) \tag{75}$$

$$H_t^{\text{hs,c,max}} = H_t^{\text{hs,d,max}} = 0.7\cdot Q^{\text{hs}} + \left(1-\gamma_t^{\text{hs}}\right)\cdot\left(0.9\cdot Q^{\text{hs}} - 0.7\cdot Q^{\text{hs}}\right) \tag{76}$$

$$\sum_{t=1}^{T}\gamma_t^{es} \geq T\cdot\varepsilon^{es} \tag{77}$$

$$\sum_{t=1}^{T}\gamma_t^{hs} \geq T\cdot\varepsilon^{hs} \tag{78}$$

Here, (75), (76), (77) and (78) demonstrate that the operational power of BESSs and HSSs can reach 0.9 times capacity with the probability of $1-\varepsilon^{es}$ and $1-\varepsilon^{hs}$, thus enhancing reliability against source-load uncertainties.

3.4. Heuristic algorithm for the solution of the bi-level scheme

As mentioned in Section 3.1, the heuristic algorithm based bi-level solution scheme is designed to solve the non-linear planning model. Its detailed iteration process is shown in Fig. 5, where the left side demonstrates the PSO algorithm and the right side shows the CEHCS expansion planning model.

3.4.1. Preliminaries of PSO algorithm

As shown in Fig. 5, the PSO algorithm is adopted to determine the configured number of servers to be added, where a certain number of particles are used to iteratively search for the optimal solution [40]. To begin with, some preliminaries are given below to better understand the PSO algorithm.

Particles. Particles are the most fundamental element of PSO, possessing five attributes, i.e., position (Q_k^{iter}), velocity (V_k^{iter}), fitness value ($fit(Q_k^{iter})$), individual optimal position ($pBest_k$), and individual optimal fitness value ($fit(pBest_k)$). Here, k and $iter$ denote the index of particles and the iteration times, respectively.

1) *Position* (Q_k^{iter}). The position of a particle represents a feasible solution of the optimization problem. That is to say, in this work, the position of a particle is a practical number of configured servers (Q^{srv}).

2) *Velocity* (V_k^{iter}). A particle moves at a certain velocity in the feasible solution space, thereby changing its position to explore more feasible solutions. In this work, the feasible solution space is the allowable range for the number of configured servers, i.e. [0, 1000], as given in Table 6.

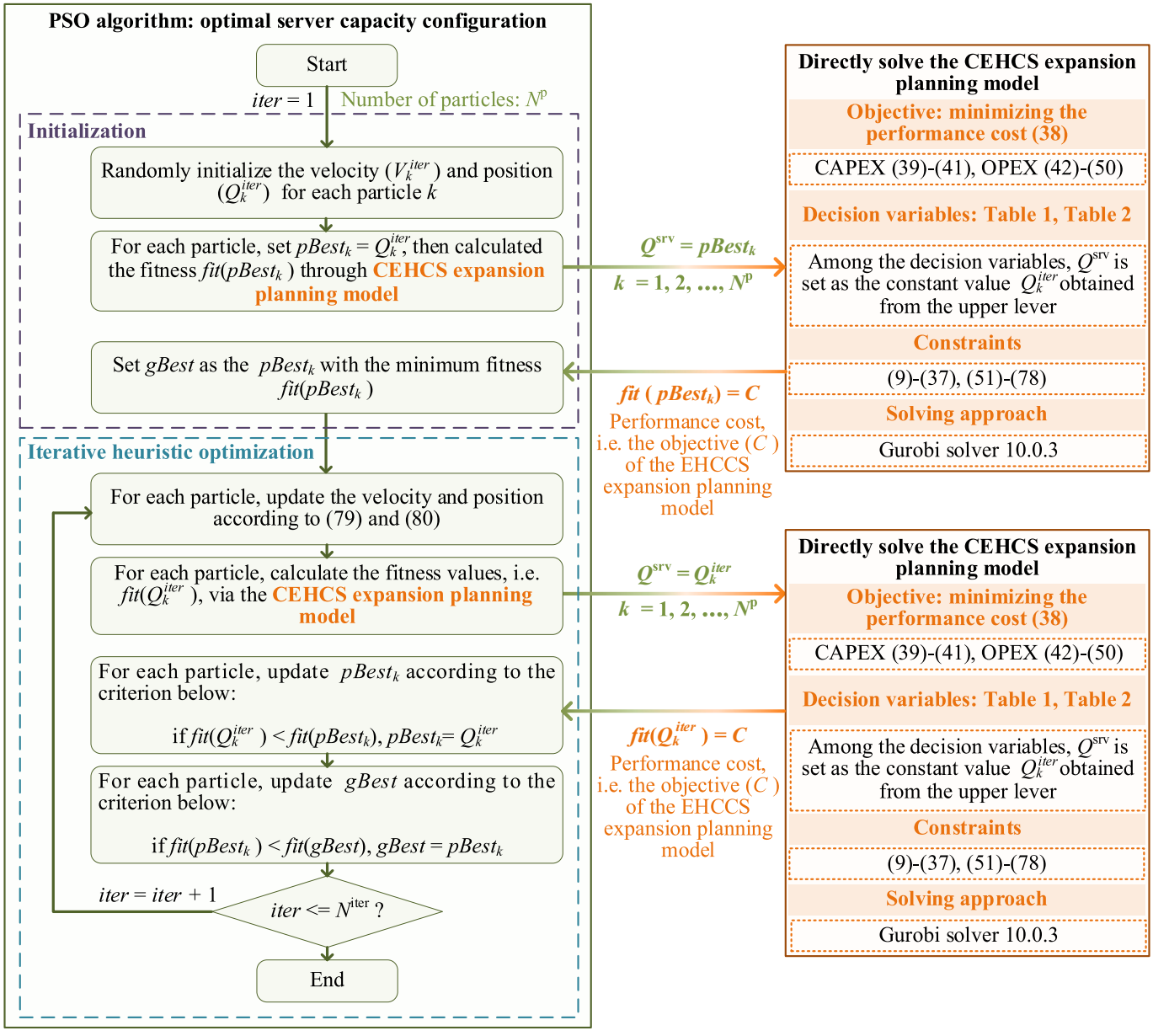


Fig. 5. Heuristic algorithm for the solution of the bi-level scheme.

3) *Fitness value* ($fit(Q_k^{iter})$). For a particle, its fitness value represents the objective value of the optimization problem under the particle's position, i.e., the performance cost (38) of the CEHCS expansion planning model under $Q^{srv} = Q_k^{iter}$.

4) *Individual optimal position* ($pBest_k$). An individual optimal position represents the position where a particle reflects its lowest fitness value.

5) *Individual optimal fitness values* ($fit(pBest_k)$). It represents the lowest fitness value found by a particle.

Particle swarm. All particles form a particle swarm, which has two attributes, i.e., global optimal fitness value ($fit(gBest)$) and global optimal position ($gBest$).

1) *Global optimal fitness value* ($fit(gBest)$). The global optimal fitness value means the lowest value among all individual optimal fitness values. It represents the performance cost of the optimal solution for the server number.

2) *Global optimal position* ($gBest$). The global optimal position represents the position where the global optimal fitness value appears, i.e., the optimal number of configured servers in this work.

It should be noted that, in our PSO implementation, both particle position and velocity are scalar (1×1) variables, as the only decision variable of the upper level is the number of servers.

3.4.2. Algorithm flow

The PSO algorithm involves two stages, i.e. initialization and iterative heuristic optimization.

Stage I: Initialization of PSO. The velocity and position of each particle are randomly initialized, and the individual optimal position of each particle is set as its initial position. Correspondingly, fitness values are calculated through solving the CEHCS expansion planning model.

Stage II: Iterative heuristic optimization of PSO. In each iteration, the velocity (V_k^{iter}) and position (Q_k^{iter}) of each particle are updated by (79) and (80), respectively. Note that the velocity (V_k^{iter}) in (79) involves three terms. The first term, i.e., ωV_k^{iter-1}, denotes the velocity of the previous iteration. The second term, i.e., $c_1 r_1 \left(pBest_k - Q_k^{iter-1}\right)$, denotes the velocity component towards the individual optimal position. And the third term, i.e., $c_2 r_2 \left(gBest - Q_k^{iter-1}\right)$ represents the velocity component towards the global optimal position. Thus, the particle velocity V_k^{iter} is affected by its own inertia, and the degree of influence can be adjusted by changing the inertia coefficient (ω). Concurrently, the second and third terms of (79) indicate that particles tend to search near individual and global optimal positions in the hope of discovering better feasible solutions. The corresponding velocity component values can be adjusted by self-learning factor (c_1), global learning factor (c_2) and random numbers (r_1 and r_2).

$$V_k^{iter} = \omega V_k^{iter-1} + c_1 r_1 \left(pBest_k - Q_k^{iter-1}\right) + c_2 r_2 \left(gBest - Q_k^{iter-1}\right) \tag{79}$$

$$Q_k^{iter} = Q_k^{iter-1} + V_k^{iter-1} \tag{80}$$

where $pBest_k$ denotes the position where particle k reflects its lowest fitness value, i.e. $fit(pBest_k)$; $gBest$ denotes the position with the lowest fitness value across all particles, i.e. $fit(gBest)$, discovered by the whole particle swarm.

Subsequently, the updated positions of particles are sent to the expansion planning model, where the number of configured servers (Q^{srv}) is set as the updated particle position (Q_k^{iter}). Further, fitness values (i.e. performance cost) are calculated by solving the CEHCS expansion model and then submitted back to the PSO flow. A more detailed process of the PSO algorithm based bi-level solution scheme can be found in Fig. 5.

Table 4
Parameters of the data center [11,38]

Parameters	Values	Parameters	Values
R^{cpu}	0.35 K/W	$Q^{cool,exist}$	100 kW
C^{cpu}	340 J/K	C^{DC}	16.19 kWh/°C
$T^{cpu,ini}$	318 K (44.85 °C)	R	1.9 °C/kW
$T^{cpu,max}$	343.15 K (70 °C)	$T^{in,min}/T^{in,max}$	15 / 25 °C
P^{idle}	0.35 kW	$r^{in,min}/r^{in,max}$	-5 / 5 °C/h
P^{peak}	0.7 kW	$t_\tau^{delay,max}$	7 h
$Q^{srv,exist}$	260		

4. Case studies

4.1. Case description and parameter settings

Test system. The case is developed from an IEEE 14-bus test power system, as illustrated in Fig. 4. The phase angle difference between adjacent nodes is constrained within [-20°, 20°].

Data source. The data profiles of renewable power generation are obtained from Xihe Energy Meteorological Big Data Platform (https://xihe-energy.com/). The electricity demand and heating load are extracted from [41], and the computational workload profiles are from the Parallel Workloads Archive [42].

Data span and resolution. Each dataset covers two years. The time horizon of a single scenario is 1 day with a 1-hour resolution.

Dataset partitioning. The first-year data (i.e. historical data, including 365 scenarios) forms the set Ω^{real}, which is applied in the forward process to obtain training sets of the denoising network. The second-year data forms the set Ω^{test}, which is used for validating the expansion planning results.

Heuristic algorithm settings. In the adopted PSO algorithm, the number of particles is set as 5, the maximum number of iterations is set as 24, the velocity coefficient (ω) is 0.5, the self-learning factor (c_1) is 0.5 and the global learning factor (c_2) is 0.6.

Planning settings. The life cycle is assumed as 10 years, and the discount rates of all equipment are set as 0.07. The unit fuel cost of CHP is 0.55 CNY/kWh and the unit heat purchase cost from DHS is 1.24 CNY/kWh, respectively. The emission factors of the utility grid, DHS and CHP, i.e. ρ^g, ρ^{dhs} and ρ^{chp}, are set as 505, 140.7 and 120 (g/kWh), respectively. The unit carbon emission costs are all set as 0.24 CNY/kg. The parameters of the data center, equipment operation, economic investment and electricity price are given in Tables 4-7, respectively.

DMOSG settings. As for the hyperparameters of the diffusion model, we set M = 2000 for all experiments and set the forward process variances to constants increasing linearly from $\beta_1 = 10^{-4}$ to $\beta_M = 0.02$. The denoising U-net model includes 2 down-sampling and 2 up-sampling processes based on 1 convolutional layer and 1 transposed convolutional layer with a kernel size of 4, a stride size of 2 and a padding size of 1, respectively. The ResNet block contains an identity mapping and two

Table 5
Operational parameters of the equipment

Equipment	Parameters	Values
Cooling system	β_1	1.05
	β_2	4 kW
WHRS	θ^{whrs}	3
	ω^{re}	0.95
CHP	ϖ	100 kW/h
	ω^{chp}	1.5
BESS	Soc^{min}	$0.1 \cdot Q^{es}$
	λ^{es}	0.001
	$\vartheta^{es,c}/\vartheta^{es,d}$	0.95 / 0.95
HSS	Soh^{min}	$0.1 \cdot Q^{hs}$
	λ^{hs}	0.001
	$\vartheta^{hs,c}/\vartheta^{hs,d}$	0.95 / 0.95

Table 6
Parameters of the capital expenditure and capacity limit

Equipment	Unit investment cost	Unit maintenance cost	Unit operational cost	Maximum capacity	Power limit
Server	8000 CNY	30	0.08 CNY	1000	/
Cooling system	1500 CNY/kW	20	0.1 CNY/kW	1000 kW	$[\beta_2, Q^{\text{cool,exist}} + Q^{\text{cool}}]$
WHRS	5400 CNY/kW	20	0.1 CNY/kW	800 kW	$[0, Q^{\text{whrs}}]$
HSS	1600 CNY/kWh	23	0.1 CNY/kWh	1500 kW	Given in (71)-(74)
CHP	6400 CNY/kW	35	0.3 CNY/kW	800 kW	$[0, Q^{\text{chp}}]$
BESS	3000 CNY/kWh	21	0.25 CNY/kWh	3000 kW	Given in (64)-(67)
WT	5000 CNY/kW	15	0.12 CNY/kW	2000 kW	Given in (56)
PV	4050 CNY/kW	5	0.08 CNY/kW	2000 kW	Given in (55)

convolutional layers followed by group normalization and the SiLU activation function. The first convolutional layer of the U-net is implemented by a kernel with a size of 3 and a padding size of 1, while the last convolutional layer is carried out by a 1×1 kernel without padding. Both of them have a stride size of 1. The model is trained through the Adam optimizer with the learning rate $lr = 10^{-4}$ and the batch size $B = 256$. The well-trained denoising network is used to generate the operational scenarios over 3650 days.

Experimental platform. The programming framework of DMOSG is PyTorch 1.11, the bi-level expansion planning model is constructed by YALMIP language in MATLAB R2021a, and the lower level of the model is solved by Gurobi 10.0.3. All experiments are implemented on the hardware platform with a 2.90-GHz Intel Core i5-10400 CPU, 48.0 GB RAM and an NVIDIA 1660 Super GPU.

4.2. Performance analysis of scenario generation

To assess the scenario generation performance of the DMOSG method, this work conducts the probabilistic distribution analysis, order moment validation and temporal correlation analysis. Meanwhile, the performance analysis also includes a direct comparison of DMOSG with three baselines, as given below.

Table 7
Purchasing price of electricity from the utility grid

Time period	Price
0:00-6:00 & 22:00-24:00	0.37 CNY/kWh
6:00-10:00 & 17:00-20:00	0.96 CNY/kWh
10:00-17:00 & 20:00-22:00	1.2 CNY/kWh

Latin hypercube sampling (LHS): LHS is a statistics-based method. It assumes that the values at each time slot follow a normal distribution, and then samples each moment separately to obtain the scenarios.

Copula function based method: Based on Copula theory, a joint probability distribution function is constructed for each time slot. The Frank-Copula function is selected to characterize the dependence between different time series. Subsequently, the joint probability distribution function is sampled at each time slot, and the sampled values are transformed inversely to generate the required scenarios.

Variational autoencoder (VAE): VAE is a learning-based method. Unlike diffusion models, the VAE does not adopt a gradual diffusion process. Instead, it directly learns the latent data distribution through its encoder-decoder architecture. The VAE model employed in this study is configured with an input dimension of 144, a hidden layer dimension of 400, and a latent space dimension of 144.

4.2.1. Probabilistic distribution analysis

Fig. 6 demonstrates the frequency distribution histogram of real data and the kernel density estimation (KDE) of both real and generated data. As a non-parametric probabilistic density estimation approach, KDE is adopted to directly estimate the distribution of generated scenarios without any prior assumptions. Overall, the histogram and kernel smoothing function (KSF) of scenarios generated by DMOSG are closest to that of real scenarios.

More specifically, due to the fact that solar power generators are unable to produce electricity when there is inadequate solar radiation, the probability density function of generated PV scenarios is very sharp around 0, but the DMOSG method has learned this feature. For other data profiles, the proposed DMOSG method also keenly captured their

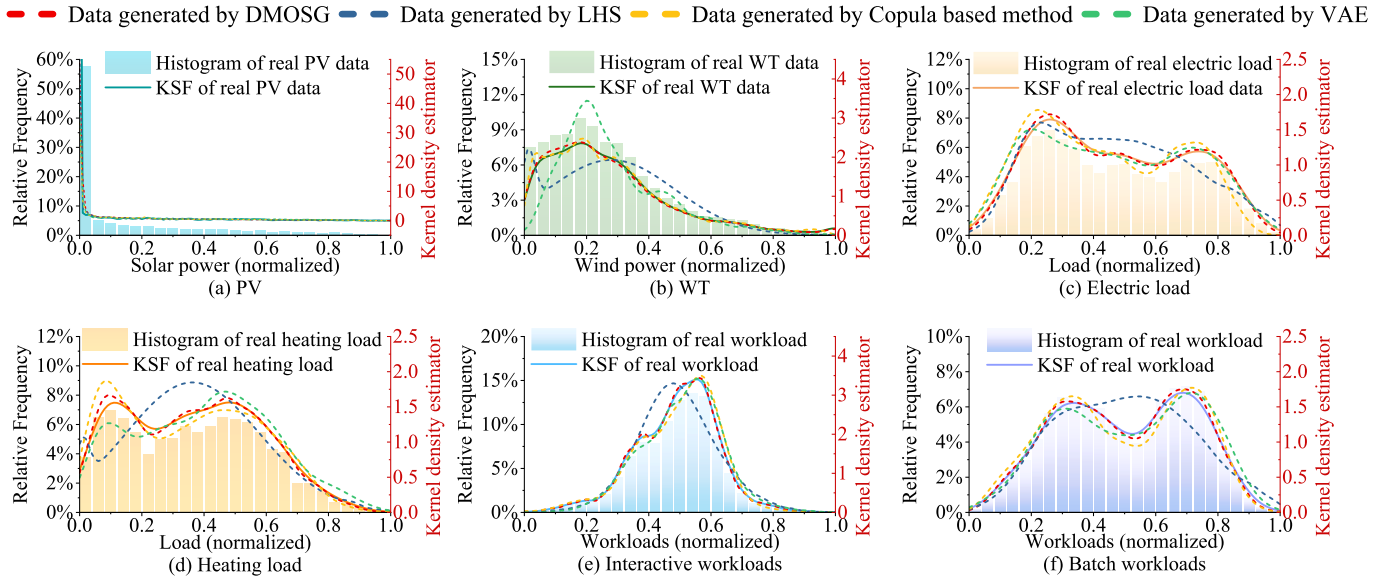


Fig. 6. Probabilistic distribution of real data and generated scenarios.

probability density distribution characteristics. The above observations indicate that the proposed DMOSG fits the probabilistic distribution of historical data well and achieves excellent posterior representation. As for the LHS, Copula, and VAE methods, they demonstrate satisfactory fitting performance for the KSF of PV data profiles. However, these baselines exhibit significant deviations on other data profiles, suggesting limited generalizability across different data distributions. Another commonly used evaluation metric is the cumulative probability distribution (CDF), as shown in Fig. 7. It's clear that the CDF errors of DMOSG are very small, within 0.025 on each data profile. Meanwhile, the errors are often larger when CDFs change rapidly, this is because the drastic changes in CDFs represent sharp distributions, which are more difficult to fit than smooth distributions. However, the CDF errors are acceptable, which indicates that the DMOSG method has well depicted the sharp parts of the probabilistic distributions, thereby representing the cumulative distribution characteristics of real data.

4.2.2. Order moment validation

Following the theory of probability and statistics, an infinite collection of order moments can uniquely define a probability distribution. Hence, through computing and comparing the multi-order moments of two distributions, it is feasible to evaluate the similarity between the generated distribution and the actual one. The first four order moments, i.e., Expectation, Variance, Skewness, and Kurtosis, are considered sufficient evaluation indicators. Therefore, the first four moments of real scenarios and generated scenarios, as well as the errors, are calculated and presented in Table 8.

Overall, the DMOSG method accurately learns the statistical features of various datasets, showing high fidelity. The order moment errors of all types of scenarios generated by DMOSG are within 5%, except for the skewness of RT and DT workloads (9.7% and 11.1% respectively). Among the scenarios generated by DMOSG, only the skewness of RT exhibits an error exceeding 10%. Nevertheless, the average relative error across all scenarios remains remarkably low at 1.9%, significantly outperforming the three baseline methods while maintaining acceptable accuracy. The order moments reflect the trends of time series changes, and the accurate fitting of them enables the model to capture the dynamic characteristics of historical observations.

4.2.3. Temporal correlation analysis

In addition to numerical statistical characteristics, temporal correlation features are also necessary metrics for assessing the quality of generated data. As random variables about time, the source-load datasets exhibit correlations between different time slots, which are depicted by autocorrelation coefficients in Fig. 8. The autocorrelation coefficient reflects the extent of correlation between any two different time slots in a time series, which is defined in (81).

$$R(\tau) = \frac{\sum_{t=1}^{T-\tau}(g_t - \mu)(g_{t+\tau} - \mu)}{\sum_{t=1}^{T}(g_t - \mu)^2} \tag{81}$$

where τ denotes the time-lag, t is the index of time slots, T denotes the length of the time series, g_t represents the value of the time sequence at time t and μ represents the expectation.

The numerical results presented in Fig. 8 are calculated by taking the average of the autocorrelation coefficients for all scenarios in a certain data profile. As can be observed, the autocorrelation of the scenarios decreases as the time-lag increases, this is because the longer time distance reduces the correlations between time slots. It can also be seen that the scenarios generated by DMOSG show almost identical temporal behaviour to the real data, with errors of less than 0.05 in PV and WT datasets, and errors of less than 0.01 in other datasets. This verifies that the DMOSG method can effectively capture the temporal characteristics of historical data.

In addition, the Copula-based method exhibits the highest error among all compared approaches. While this method can capture cross-variable dependencies, it fails to adequately model temporal autocorrelations in individual time series. Although both LHS and VAE demonstrate better performance over the Copula-based method in fitting temporal behaviour, their errors remain larger than those of DMOSG.

4.3. Ablation study of hyperparameters

To help improve the applicability to various datasets for the proposed scenario generation solution, the impact of diffusion steps (M) on the diffusion model is explored in this work. The evolutions of the loss during training for different numbers of diffusion steps are demonstrated in Fig. 9. One observation is that the fitting loss of the denoising network

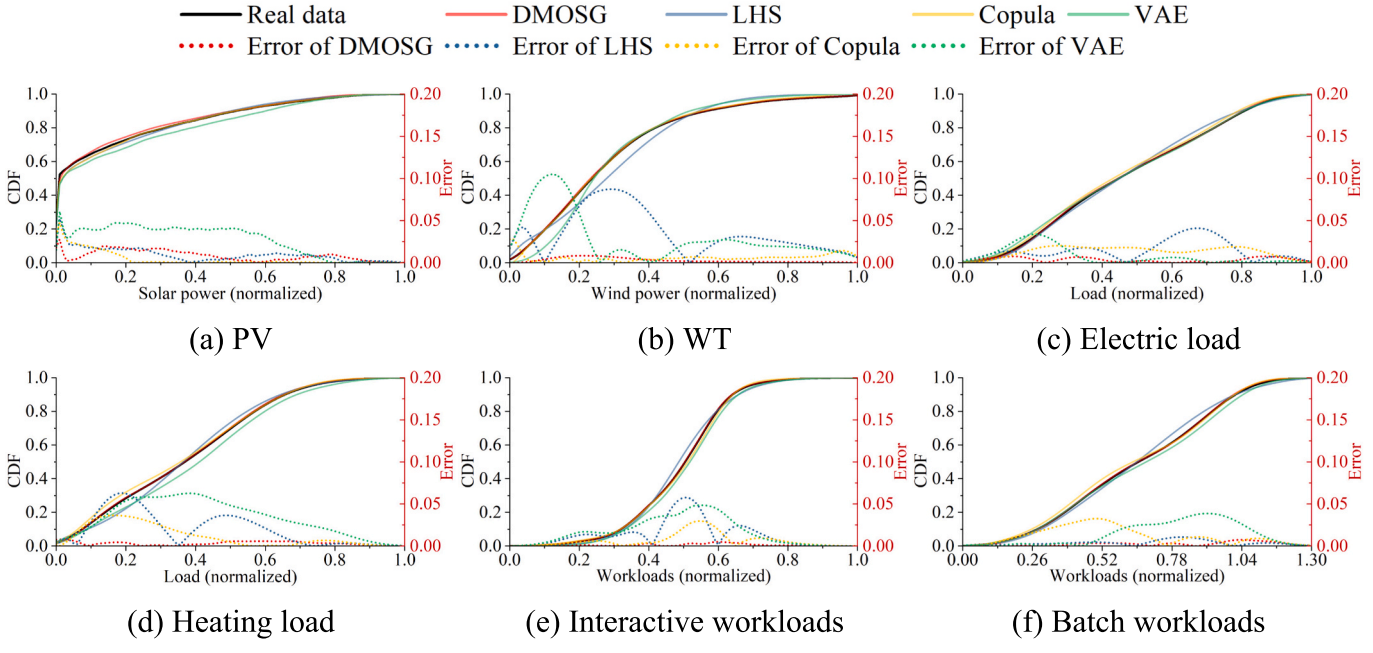


Fig. 7. Cumulative distribution functions of the real data and generated scenarios.

Table 8
Order moment validation

Scenario type	Statistic features	Real data	DMOSG	LHS	Copula	VAE
PV	Expectation	0.1456	0.1457 (0.1%)	0.1492 (2.4%)	0.1407 (-3.4%)	0.1725 (18.5%)
	Variance	0.0507	0.0511 (0.7%)	0.048 (-5.3%)	0.0471 (-7.1%)	0.0599 (18.2%)
	Skewness	1.5950	1.5328 (-3.9%)	1.5293 (-4.1%)	1.6428 (3.0%)	1.3058 (-18.1%)
	Kurtosis	4.5537	4.4442 (-2.4%)	4.4902 (-1.4%)	4.6243 (1.6%)	3.4533 (-24.2%)
WT	Expectation	0.2782	0.2771 (-0.4%)	0.2878 (3.4%)	0.2446 (-12.1%)	0.2836 (1.9%)
	Variance	0.0429	0.0415 (-3.3%)	0.0359 (-16.4%)	0.0376 (-12.4%)	0.0282 (-34.2%)
	Skewness	1.2315	1.2033 (-2.3%)	0.3946 (-68%)	1.3039 (5.9%)	1.1507 (-6.6%)
	Kurtosis	4.5313	4.5014 (-0.7%)	2.6557 (-41.4%)	4.3801 (-3.3%)	4.2974 (-5.2%)
Electric load	Expectation	0.4692	0.4673 (-0.4%)	0.4665 (-0.6%)	0.4431 (-5.6%)	0.4637 (-1.2%)
	Variance	0.0558	0.0562 (0.7%)	0.0527 (-5.5%)	0.0602 (7.9%)	0.0593 (6.3%)
	Skewness	0.1807	0.1792 (-0.8%)	0.2706 (49.8%)	0.1922 (6.4%)	0.1377 (-23.8%)
	Kurtosis	1.8415	1.8353 (-0.3%)	2.1291 (15.6%)	1.8211 (-1.1%)	1.7991 (-2.3%)
Heating load	Expectation	0.3660	0.3642 (-0.5%)	0.3696 (1.0%)	0.3504 (-4.3%)	0.3998 (9.2%)
	Variance	0.0465	0.0461 (-0.8%)	0.0424 (-8.7%)	0.0448 (-3.7%)	0.0488 (4.9%)
	Skewness	0.1556	0.1494 (-4.0%)	0.2129 (36.8%)	0.1640 (5.4%)	0.0854 (-45.1%)
	Kurtosis	2.0847	2.0771 (-0.4%)	2.5854 (24%)	1.9047 (-8.6%)	2.2120 (6.1%)
RT workloads	Expectation	0.4904	0.4892 (-0.2%)	0.4902 (0.0%)	0.4863 (-0.8%)	0.5074 (3.5%)
	Variance	0.0173	0.0174 (0.6%)	0.0169 (-2.2%)	0.0156 (-9.8%)	0.0164 (-5.0%)
	Skewness	-0.3299	-0.3620 (9.7%)	0.1177 (-135.7%)	-0.3452 (4.6%)	-0.1387 (-58%)
	Kurtosis	3.5635	3.5310 (-0.9%)	3.3610 (-5.7%)	3.5576 (-0.2%)	3.2398 (-9.1%)
DT workloads	Expectation	0.5019	0.5006 (-0.3%)	0.5011 (-0.2%)	0.5096 (1.5%)	0.5152 (2.6%)
	Variance	0.0450	0.0452 (0.4%)	0.0435 (-3.4%)	0.0442 (-1.8%)	0.0486 (8.0%)
	Skewness	-0.0778	-0.0864 (11.1%)	0.0924 (-218.8%)	-0.0712 (-8.5%)	-0.1180 (51.7%)
	Kurtosis	1.9237	1.9175 (-0.3%)	2.3091 (20.0%)	1.8291 (-4.9%)	1.8559 (-3.5%)
Average relative error			**1.9%**	**27.9%**	**5.2%**	**15.3%**

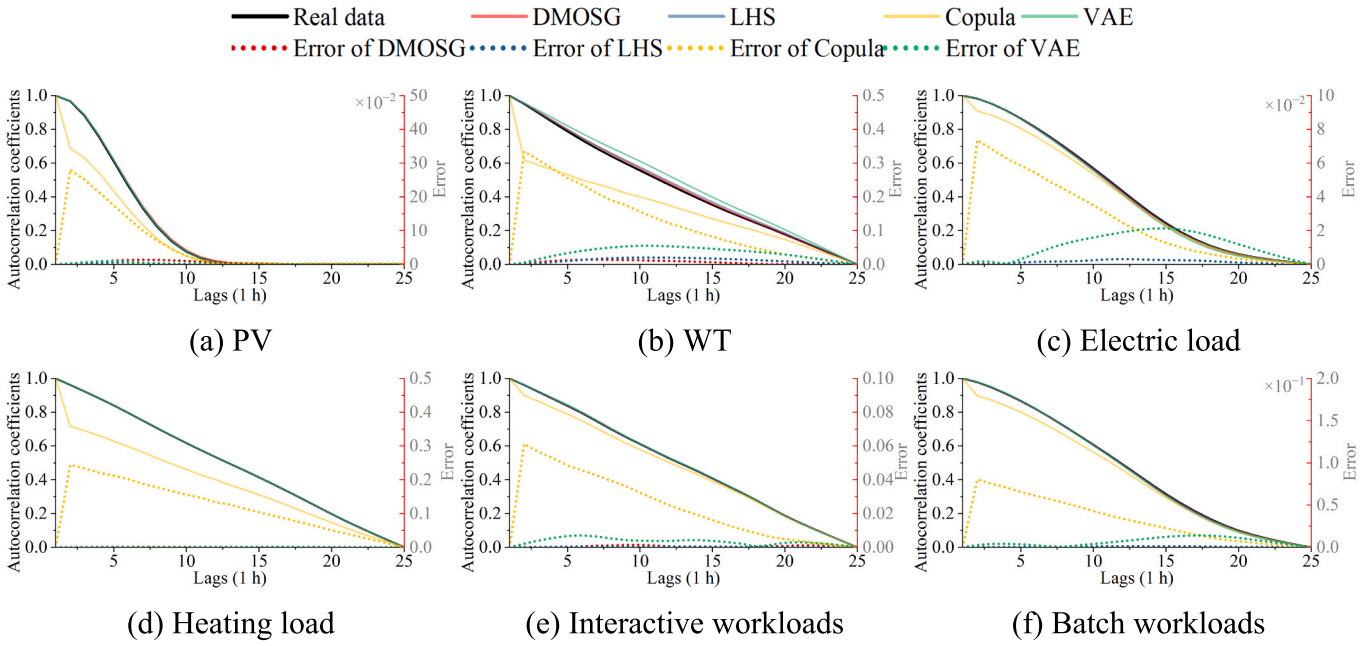


Fig. 8. Autocorrelation of real data and generated scenarios.

decreases as the number of steps (M) increases. The denoising process can be likened to carving, where more steps bring more delicate knife techniques, making the statue more realistic. In the experiment, we find that ideal results can be obtained when the number of steps reaches 2000. At this point, increasing the number of steps does not significantly improve performance, while the computational cost increases drastically. Therefore, the number of diffusion steps (M) is set to 2000 in this work.

Another hyper-parameter to be studied is forward process variance, as shown in Fig. 10. The increase of β_M induces loss reduction, and the performance improvement is no longer significant when β_M reaches 0.02. The reverse generation process starts from the Gaussian distribution, the large value of β_M makes $\mathbf{x}_M$ close to the Gaussian distribution,

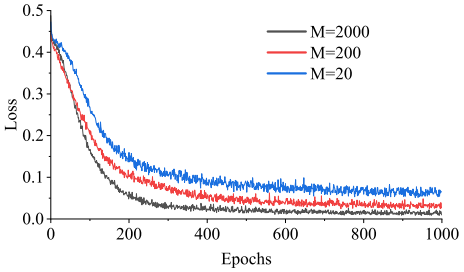


Fig. 9. Losses during training under different diffusion steps.

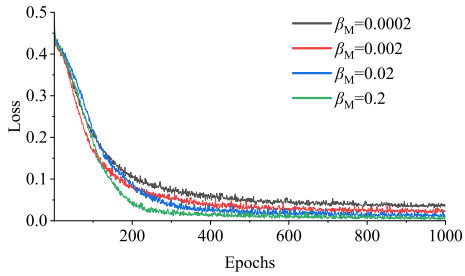


Fig. 10. Losses during training under different β_M.

thus improving the model performance. However, a too large value of β_M disrupts the initial distribution pattern of the data, resulting in the incapacity of further improvement for model accuracy.

4.4. Analysis of the planning strategy

In this work, the DMOSG is used to generate 3650 scenarios, and the k-medoids algorithm is adopted to obtain 600 representative scenarios. The set Ω^{ge} denotes the 600 generated representative scenarios. Meanwhile, as described in Section 4.1, Ω^{real} denotes the historical scenarios (365 scenarios).

To verify the effectiveness of the proposed data-driven scenario generation based bi-level expansion planning approach, this work considers 14 various schemes for comparative analysis, as shown in Table 9.

Scheme #1: (**the proposed solution**): The DMOSG and bi-level expansion planning model are both adopted for DTC infrastructure upgrading.

Scheme #2: The bi-level expansion planning model is applied, while only historical data is used for capacity configuration, with DMOSG not adopted.

Scheme #3 (**no CEHCS upgrading**): the DMOSG is used, but only servers and cooling systems are considered in the expansion planning.

Further, to illustrate the role and necessity of different devices in the proposed planning scheme, some components are considered separately in **Schemes #4 - #14**.

4.4.1. Planning results and economic analysis

(1) Capacity configuration results

The expansion planning results of different schemes are given in Table 10. The planning results reveal that Scheme #1 demonstrates more configured capacity across nearly all devices relative to Scheme #2. This enhancement stems from the incorporation of potential scenarios generated by DMOSG in Scheme #1. While these potential scenarios are not included in the historical observations, they may occur in the future. Consequently, Scheme #1 configures more equipment capacity to enhance system reliability and maintain high energy efficiency performance across all potential future scenarios.

In contrast to Scheme #1, Scheme #4 exhibits a 56.1 kW increase in WHRS capacity. This is because HSS and CHP are not considered in Scheme #4, and hence, more capacity of WHRS is needed to supply the heating load. In Scheme #5, the HSS configuration capacity is 0, demonstrating that thermal energy storage becomes uneconomical when solely relying on heat purchased from DHS. This limitation arises from two reasons: a) energy loss exists during the heat absorption and release cycle; and b) the price of heat purchased from DHS is fixed, with no difference or profit margin to earn. However, as evidenced in Scheme #6, Scheme #8, and Scheme #9, the presence of either WHRS or CHP enables effective utilization of HSS to address temporal mismatches between heat generation and demand by storing surplus produced thermal energy.

The observations of Schemes #10 - #14 reveal some roles of BESS and RES. In contrast to the unused HSS in Scheme #5, Scheme #10 demonstrates that BESS alone can help achieve economic benefits. This cost reduction is enabled by time-of-use electricity pricing from the grid, which allows BESS to charge during low-price periods and discharge during peak-price periods, effectively taking the opportunity of electricity price arbitrage. Further, Scheme #13 demonstrates a 331.3 kW

Table 9
Different schemes for comparative analysis

	Description	Scenarios used for planning	Planned devices							
			Server	Cooling system	WHRS	HSS	CHP	BESS	WT	PV
Scheme #1	Proposed CEHCS upgrading with DMOSG	Ω^{ge}	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Scheme #2	CEHCS upgrading using historical scenarios	Ω^{real}	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Scheme #3	No CEHCS upgrading but DMOSG is applied	Ω^{ge}	Yes	Yes	/	/	/	/	/	/
Scheme #4	In Schemes #4 - #14, CEHCS is applied, but the planned equipment is considered separately, thus demonstrating how much each component contributes to total investment and operating costs.	Ω^{ge}	Yes	Yes	Yes	/	/	/	/	/
Scheme #5		Ω^{ge}	Yes	Yes	/	Yes	/	/	/	/
Scheme #6		Ω^{ge}	Yes	Yes	Yes	Yes	/	/	/	/
Scheme #7		Ω^{ge}	Yes	Yes	/	/	Yes	/	/	/
Scheme #8		Ω^{ge}	Yes	Yes	/	Yes	Yes	/	/	/
Scheme #9		Ω^{ge}	Yes	Yes	Yes	Yes	Yes	/	/	/
Scheme #10		Ω^{ge}	Yes	Yes	/	/	/	Yes	/	/
Scheme #11		Ω^{ge}	Yes	Yes	/	/	/	/	Yes	/
Scheme #12		Ω^{ge}	Yes	Yes	/	/	/	/	/	Yes
Scheme #13		Ω^{ge}	Yes	Yes	/	/	/	/	Yes	Yes
Scheme #14		Ω^{ge}	Yes	Yes	/	/	/	Yes	Yes	Yes

Table 10
Expansion planning results of equipment capacity under different schemes

	servers	cooling system (kW)	WHRS (kW)	HSS (kWh)	CHP (kW)	BESS (kWh)	WT (kW)	PV (kW)
Scheme #1	660	561.0	96.3	157.5	381.7	514.5	678.5	1382.9
Scheme #2	577	501.7	90.4	149.7	376.3	259.3	850.7	1054.5
Scheme #3	706	594.5	/	/	/	/	/	/
Scheme #4	700	589.8	152.4	/	/	/	/	/
Scheme #5	706	594.5	/	0	/	/	/	/
Scheme #6	700	589.8	152.4	108.6	/	/	/	/
Scheme #7	706	594.5	/	/	432.6	/	/	/
Scheme #8	706	594.5	/	1234.0	446.6	/	/	/
Scheme #9	706	594.5	49.7	1109.7	437.9	/	/	/
Scheme #10	706	594.5	/	/	/	2836.5	/	/
Scheme #11	702	591.5	/	/	/	/	1235.7	/
Scheme #12	700	589.8	/	/	/	/	/	1589.4
Scheme #13	700	589.8	/	/	/	/	904.4	1298.4
Scheme #14	680	575.4	/	/	/	1310.3	848.0	1460.3

reduction in WT capacity compared to Scheme #11, coupled with a 291 kW decrease in PV capacity relative to Scheme #12. This capacity reduction confirms the operational complementarity between WT and PV, where their combined generation patterns mitigate individual power supply requirements.

(2) Breakdown of capital and operating expenditure

Table 11 presents the breakdown of CAPEX and OPEX by equipment type under Schemes #1 - #3. The CAPEX includes investment cost and maintenance cost. Other expenses are classified as OPEX.

Scheme #1 requires a CAPEX of 3.608 million CNY, representing a 11.6% increase (0.375 million CNY) compared to the CAPEX of Scheme #2's 3.233 million CNY. However, this higher initial investment is offset by better operational efficiency, as Scheme #1's OPEX of 28.0 million CNY represents an 11.1% (3.5 million CNY) reduction compared to Scheme #2's OPEX of 31.5 million CNY. Consequently, the total cost of Scheme #1 is 3.2 million CNY (9.2%) lower than Scheme #2 over the lifecycle, validating the economic advantage of the proposed solution. More specifically, Scheme #1 employs the generated scenario set (Ω^{ge}) for planning, in contrast to Scheme #2, which relies exclusively on historical scenarios (Ω^{real}). Here, Ω^{ge} incorporates potential future scenarios that are not observed in Ω^{real} but may realistically occur. Consequently, the higher CAPEX in Scheme #1 can be interpreted as a strategic investment in future risk resilience, which helps maintain operational efficiency (i.e., low OPEX) across a wider range of possible future scenarios.

Scheme #3 has a CAPEX of 1.262 million CNY, which is significantly lower than that of Scheme #1 (3.608 million CNY) by 2.346 million CNY due to the absence of the CEHCS upgrading. However, the OPEX of Scheme #3 reaches 71.9 million CNY, approximately 43.9 million CNY higher than Scheme #1's OPEX (28.0 million CNY). Consequently, the total cost of Scheme #1 is 41.6 million CNY lower than Scheme #3, representing a substantial 56.8% reduction. These findings demonstrate that while the CEHCS upgrading requires a higher initial investment, it yields considerable long-term benefits by dramatically reducing operational costs and overall expenditures.

To demonstrate how much each component contributes to total investment and operating costs, Table 12 lists the CAPEX and OPEX of Schemes #3 - #14. Scheme #3 (no CEHCS upgrading) is considered a baseline. Below is a separate analysis of the contribution of each device to performance cost.

WHRS. The CAPEX of Scheme #4 (with only WHRS integration) amounts to 1.4 million CNY, seeing a rise by 0.14 million CNY (11.1%) compared to Scheme #3. However, the OPEX is reduced to 66.1 million

Table 11
Breakdown of CAPEX and OPEX by equipment type under Scheme #1 - #3

Scheme	Device	Investment cost (CNY)	Maintenance cost (CNY)	Penalty for QoS not met (CNY)	Penalty for RES overproduction (CNY)	Operating cost (CNY)	Fuel cost (CNY)	Heating purchase cost (CNY)	Carbon emission cost (CNY)	Power purchase cost (CNY)	Total of scheme (CNY)
#1	Server	7.52×10^5	1.98×10^5	0	\	1.61×10^6	\	2.33×10^6	1.39×10^6	5.43×10^6	**3.16×10^7** (↓56.8%)
	Cooling system	1.20×10^5	1.12×10^5	\	\	2.10×10^6	\				
	WHRS	7.41×10^4	1.93×10^4	\	\	2.46×10^5	\				
	HSS	3.59×10^4	3.62×10^4		\	8.06×10^4	\				
	CHP	3.48×10^5	1.34×10^5	\	\	3.25×10^6	5.95×10^6				
	BESS	2.20×10^5	1.08×10^5	\	\	5.60×10^5	\				
	WT	4.83×10^5	1.02×10^5		1.04×10^6	1.79×10^6					
	PV	7.97×10^5	6.91×10^4	\	1.21×10^6	1.01×10^6	\				
	Total	**2.83×10^6**	**7.78×10^5**	**0**	**2.25×10^6**	**1.06×10^7**	**5.95×10^6**	**2.33×10^6**	**1.39×10^6**	**5.43×10^6**	
#2	Server	6.58×10^5	1.73×10^5	0	\	1.65×10^6	\	3.24×10^6	1.76×10^6	8.94×10^6	**3.48×10^7** (↓52.5%)
	Cooling system	1.07×10^5	1.00×10^5	\	\	2.16×10^6	\				
	WHRS	6.95×10^4	1.81×10^4	\	\	1.97×10^5	\				
	HSS	3.41×10^4	3.44×10^4		\	6.98×10^4	\				
	CHP	3.43×10^5	1.32×10^5	\	\	3.04×10^6	5.57×10^6				
	BESS	1.11×10^5	5.45×10^4	\	\	3.31×10^5	\				
	WT	6.06×10^5	1.28×10^5		1.83×10^6	1.72×10^6	\				
	PV	6.08×10^5	5.27×10^4	\	2.02×10^5	8.20×10^5					
	Total	**2.54×10^6**	**6.93×10^5**	**0**	**2.03×10^6**	**9.98×10^6**	**5.57×10^6**	**3.24×10^6**	**1.76×10^6**	**8.94×10^6**	
#3	Server	8.05×10^5	2.12×10^5	0	\	1.65×10^6	\	2.06×10^7	5.24×10^6	4.23×10^7	**7.32×10^7** (base)
	Cooling system	1.27×10^5	1.19×10^5	\	\	2.14×10^6	\				
	Total	**9.32×10^5**	**3.31×10^5**	**0**	**0**	**3.79×10^6**	**0**	**2.06×10^7**	**5.24×10^6**	**4.23×10^7**	

Table 12
CAPEX and OPEX of Scheme #3 - #14

Scheme	CAPEX (CNY)	Penalty for RES overproduction (CNY)	Operating cost (CNY)	Fuel cost (CNY)	Heating purchase cost (CNY)	Carbon emission cost (CNY)	Power purchase cost (CNY)	Total cost of the scheme (CNY)
#3	1.26×10^6	0	3.79×10^6	0	2.06×10^7(base)	5.24×10^6 (base)	4.23×10^7 (base)	**7.32×10^7 (base)**
#4	1.40×10^6	0	4.16×10^6	0	1.15×10^7 (↓44.2%)	5.18×10^6 (↓1.2%)	4.53×10^7 (↑7.1%)	**6.75×10^7 (↓7.8%)**
#5	1.26×10^6	0	3.79×10^6	0	2.06×10^7 (↓0.0%)	5.24×10^6 (↓0.0%)	4.23×10^7 (↓0.0%)	**7.32×10^7(↓0.0%)**
#6	1.45×10^6	0	4.22×10^6	0	1.14×10^7 (↓44.8%)	5.18×10^6 (↓1.2%)	4.53×10^7 (↑7.1%)	**6.748×10^7(↓7.8%)**
#7	1.81×10^6	0	8.68×10^6	8.95×10^6	4.52×10^5 (↓97.8%)	3.50×10^6 (↓33.2%)	2.85×10^7 (↓32.7%)	**5.18×10^7(↓29.2%)**
#8	2.39×10^6	0	9.38×10^6	9.14×10^6	3.34×10^5 (↓98.4%)	3.47×10^6 (↓33.7%)	2.67×10^7 (↓36.9%)	**5.14×10^7(↓29.7%)**
#9	2.37×10^6	0	9.03×10^6	8.44×10^6	4.47×10^4 (↓99.8%)	3.60×10^6 (↓31.3%)	2.76×10^7 (↓34.8%)	**5.11×10^7(↓30.2%)**
#10	3.07×10^6	0	8.44×10^6	0	2.06×10^7 (↓0.0%)	5.33×10^6 (↑1.7%)	3.55×10^7 (↓16.1%)	**7.29×10^7(↓0.4%)**
#11	2.32×10^6	3.26×10^6	6.24×10^6	0	2.06×10^7 (↓0.0%)	3.52×10^6 (↓32.9%)	2.56×10^7 (↓39.4%)	**6.16×10^7(↓15.9%)**
#12	2.25×10^6	2.32×10^6	4.93×10^6	0	2.06×10^7 (↓0.0%)	3.63×10^6 (↓30.6%)	2.31×10^7 (↓45.4%)	**5.68×10^7(↓22.4%)**
#13	2.84×10^6	4.11×10^6	6.49×10^6	0	2.06×10^7 (↓0.0%)	2.66×10^6 (↓49.3%)	1.46×10^7 (↓65.5%)	**5.13×10^7(↓29.9%)**
#14	3.70×10^6	3.68×10^6	8.31×10^6	0	2.06×10^7 (↓0.0%)	2.57×10^6 (↓51%)	1.18×10^7 (↓72%)	**5.07×10^7(↓30.7%)**

Note: As the punishment cost for QoS not met remains 0 across all schemes, it has been omitted from this table presentation.

CNY, representing a significant fall of 5.8 million CNY (8.1%) relative to Scheme #3. Thus, the total cost of Scheme #4 demonstrates an overall reduction of approximately 5.66 million CNY (7.77%) compared to Scheme #3. More specifically, in contrast to Scheme #3, Scheme #4 achieves significant cost reductions in heat procurement and carbon emission by 9.1 million CNY (i.e. 44.2%) and 0.06 million CNY (i.e. 1.2%), respectively. This implies that the WHRS enhances the system's economic and environmental performance by reducing the dependence on DHS through the utilization of waste heat.

CHP. The comparison between Scheme #7 and Scheme #3 shows that while the CHP incurs an additional fuel cost of 8.95 million CNY, it achieves significant reductions in heating purchase cost (97.8%), carbon emission cost (33.2%) and power purchase cost (32.7%), ultimately reaching a 29.2% reduction in total cost.

HSS. While HSS alone (Scheme #5) demonstrates no cost-saving potential, its synergistic combination with WHRS or CHP (Schemes #6, #8, and #9) yields measurable economic advantages. For example, in contrast to Scheme #7 (with only CHP), Scheme #8 (with CHP and HSS) achieves reductions of 0.118 million CNY (26%) in heat purchase cost, 0.03 million CNY (0.9%) in carbon emission cost, and 1.8 million CNY (6.3%) in electricity expenditures, culminating in an overall cost reduction of 0.4 million CNY.

WT and PV. It can be observed from Scheme #11 (with only WT) and Scheme #12 (with only PV) that RES helps improve economic efficiency by reducing carbon emissions and the power purchased from the utility grid. Besides, Scheme #13 (with both WT and PV) implies that the combination of PV and WT effectively capitalizes on their temporal complementarity, thereby achieving a greater reduction in total costs.

BESS. Scheme #10 (with only BESS) indicates that BESS contributes to total costs by reducing power purchase expenditure. Meanwhile, the comparison between Scheme #13 (with WT and PV) and Scheme #14 (with BESS, WT and PV) reveals that the integration of BESS and RES is more energy-efficient than using them separately. This is because BESS can effectively address the temporal mismatch between RES generation and load demand by charging during high RES-production periods and discharging during peak demand periods.

The above analysis reveals how each device contributes to the total cost. Also, the integration of multiple energy devices is confirmed to be essential in achieving optimal system performance, as it enables the complementarity among different equipment.

4.4.2. Energy efficiency analysis

Power usage effectiveness (PUE) is adopted to assess the energy efficiency of DTCs in this work, as defined in (82). A lower PUE value represents higher energy efficiency. It means that all energy is consumed by servers when $PUE = 1$.

$$PUE = \sum_{t=1}^{T} \left(P_t^{\mathrm{IT}} + P_t^{\mathrm{cool}} + P_t^{\mathrm{whrs}} - H_t^{\mathrm{re}}\right) \Big/ \sum_{t=1}^{T} P_t^{\mathrm{srv}} \tag{82}$$

Fig. 11 illustrates the PUE values of 40 test days (which are randomly selected from Ω^{test}) under 14 schemes. It is clear that Scheme #1 consistently demonstrates lower PUE values than Scheme #3 (with no CEHCS upgrading) across all test days. The most significant performance gap is observed on the 6th test day, where Scheme #1 achieves a PUE of 1.591, showing a 0.45 reduction compared to Scheme #3 (2.041). The smallest gap occurs on the 26th test day, with Scheme #1's PUE (2.033) being 0.008 lower than that of Scheme #3 (2.041). The above observations indicate that CEHCS upgrading significantly enhances the energy efficiency of DTCs. Additionally, although Schemes #4 and #6 also achieve low PUE values, their notably higher OPEX (as quantified in Section 4.5.1) make them not the most cost-effective solution.

4.4.3. Operation analysis

Fig. 12 illustrates the dispatching results of the DC i.e., the power balance, the computational workload processing, the indoor temperature and the average CPU temperature, in a representative day under S#1. The cooling power and server heat dissipation exhibit a generally consistent variation trend, with subtle discrepancies attributed to the flexibility of the temperature management and cooling power adjustment. The average CPU temperature is significantly affected by the power consumption of servers (i.e., server-dissipated heat), rising to a peak of approximately 74 °C at the 23rd time slot. The observation that the average CPU temperature remains consistently within the safe operating limit demonstrates the effectiveness of the proposed thermal management strategy.

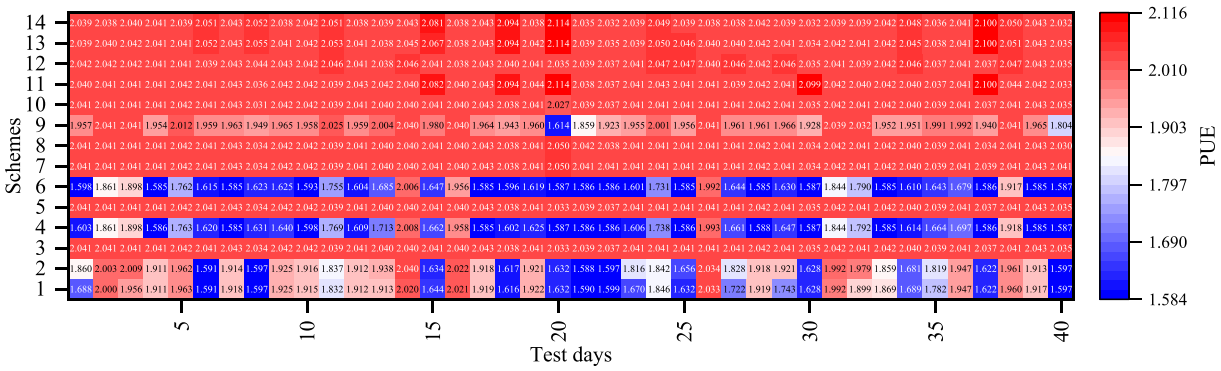


Fig. 11. PUE values of 14 schemes under 40 test days.

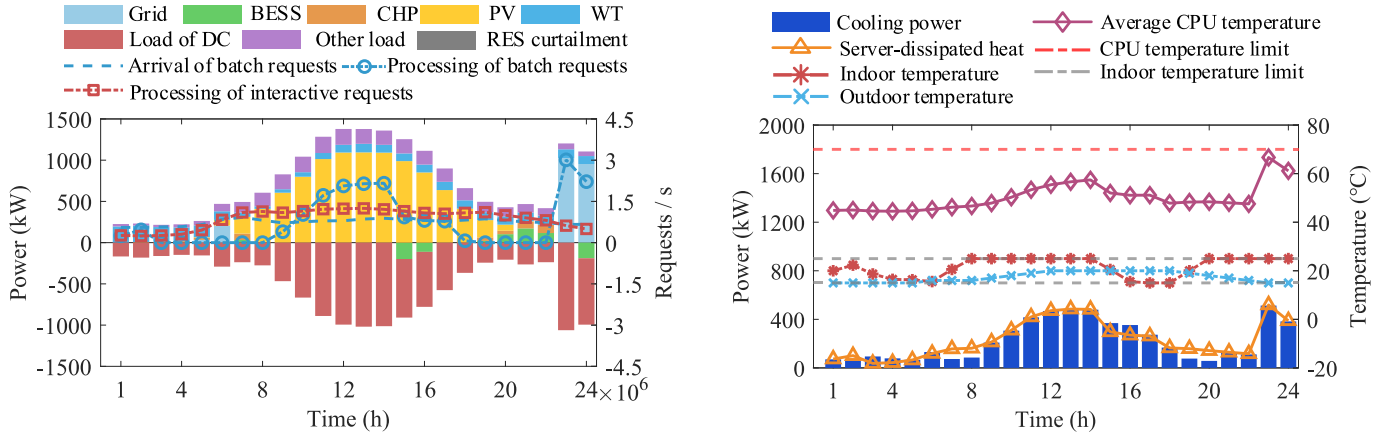


Fig. 12. Dispatching analysis considering CPU temperature and workload processing under S#1.

4.5. Sensitivity analysis

4.5.1. Sensitivity to electricity price

Fig. 13 illustrates the sensitivity of equipment capacity to electricity price under Schemes #1, #2 and #3. The capacities of BESS and HSS are very sensitive to electricity price, with a significant positive correlation. In Scheme #1, an 80% increase in electricity price triggers capacity expansions of 318.3% for BESS and 393.6% for HSS, respectively. Scheme #2 demonstrates similar trends, with BESS and HSS capacities rising by 428.5% and 186.1%, respectively, under the same electricity price escalation. This observation can be attributed to the amplified peak-to-valley price differentials caused by rising electricity prices, which enable BESSs to save more on electricity bills.

Fig. 14 illustrates the sensitivity of performance cost to electricity price. The total cost of Scheme #3 is the most sensitive to the electricity price, seeing a large rise of 46.2% when the electricity price increases by 80%. Nevertheless, Scheme #1 shows the highest stability, with total cost consistently lower than Scheme #2 and #3. Fig. 15 illustrates the sensitivity of PUE to electricity price. The PUE of Scheme #3 is not affected by electricity prices. When the electricity price increases by 40%, the PUEs of Schemes #1 and #2 rise to a peak of 1.861 and 1.929, respectively, seeing slight decreases after that.

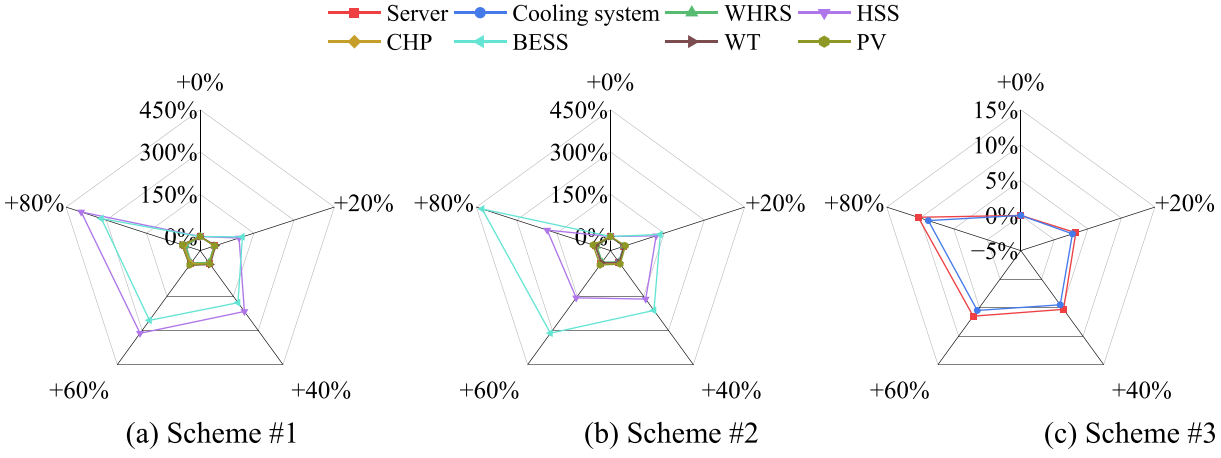


Fig. 13. Sensitivity of equipment capacity to electricity price.

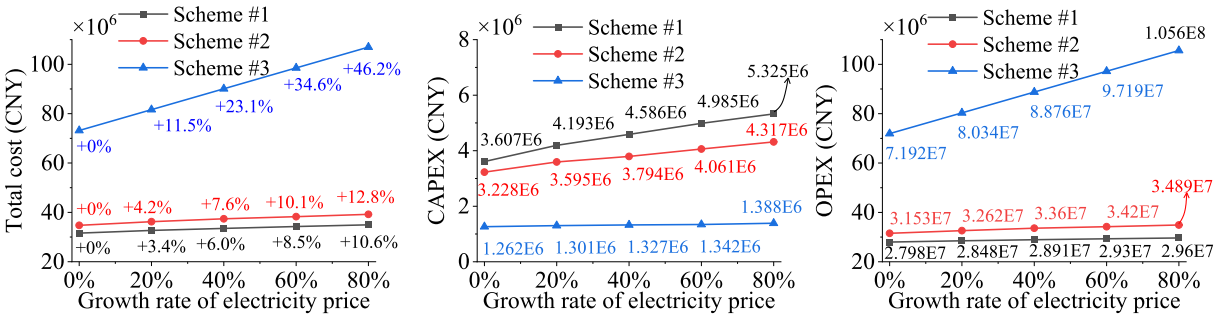


Fig. 14. Sensitivity of performance cost to electricity price.

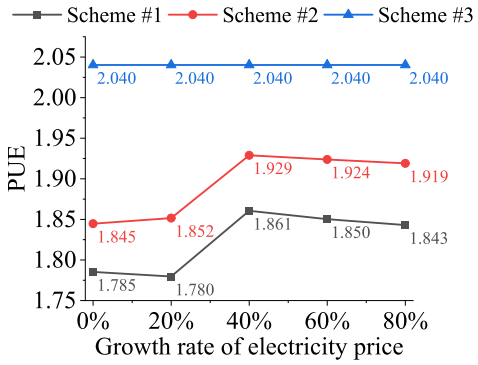


Fig. 15. Sensitivity of PUE to electricity price.

4.5.2. Sensitivity to fuel price

Fig. 16 demonstrates the sensitivity of equipment capacity to fuel prices, respectively. It is clear in Schemes #1 and #2 that the configured capacity of BESS increases significantly as the fuel price rises. In Scheme #1, the fuel price escalation of 20%, 40%, 60% and 80% makes BESS capacity rise by 20.1%, 21.7%, 39.0% and 77.1% respectively. While in Scheme #2, the BESS capacity sees rises by 47.6%, 70.3%, 153% and 185.8% respectively. Besides, WHRS capacity is also sensitive to fuel prices, with increases of 4.3%, 5.2%, 7.4% and 21.8% in Scheme #1 and 7.4%, 13.5%, 25.8% and 38.9% in Scheme #2. This is because the rise in fuel prices leads to an increase in the cost of electricity and heat production for CHP units, leading CEHCS to plan more energy storage and WHRS to reduce its dependence on CHP.

Fig. 17 demonstrates the sensitivity of performance cost to fuel prices. It can be observed that Scheme #1 exhibits the lowest cost under all fuel prices. When the fuel price increases by 80%, the total cost of scheme #1 only increases by 12.9%. Fig. 18 demonstrates the sensitivity of PUE to fuel prices. As fuel price changes, Scheme #1 consistently maintains a lower PUE than other schemes, demonstrating higher energy efficiency.

4.5.3. Sensitivity of the scenario number

Since the proposed planning solution is based on scenario generation, the number of clustered generated scenarios may have a significant impact on the results in Scheme #1. Thus, we evaluate the sensitivity of numerical results to the number of scenarios. It is noted that Schemes #2 and #3 are not affected by generated scenarios and are only used as benchmarks to assess Scheme #1 in this Section.

Fig. 19 illustrates the sensitivity of equipment capacity to scenario number in Scheme #1. Due to the fact that wind power usually exhibits the strongest volatility and randomness, it is most sensitive to the number of scenarios. A limited number of scenarios cannot adequately capture the uncertainty of wind power. Consequently, the configured capacity of WTs tends to increase as the number of scenarios decreases, to hedge against potential uncertainties.

Fig. 20 demonstrates the sensitivity of performance cost to scenario number in Scheme #1. It is shown that the total cost sees a significant rise by 31.8% when the scenario number decreases to 280. This is because too few scenarios cannot effectively depict the probabilistic distribution of real data, resulting in planning results that cannot fully address uncertainties in the test set (Ω^{test}). Although the performance cost increases dramatically when the number of clustered scenarios falls below the number of historical scenarios, Scheme #1 still performs better than Scheme #3 in total cost. The observations also inspire us that the premise for Scheme #1 to be better than Scheme #2 is the sufficient number of clustered generated scenarios. Besides, Fig. 21 demonstrates that Scheme #1 maintains better PUE than the other two comparative schemes under different numbers of clustered scenarios, confirming the

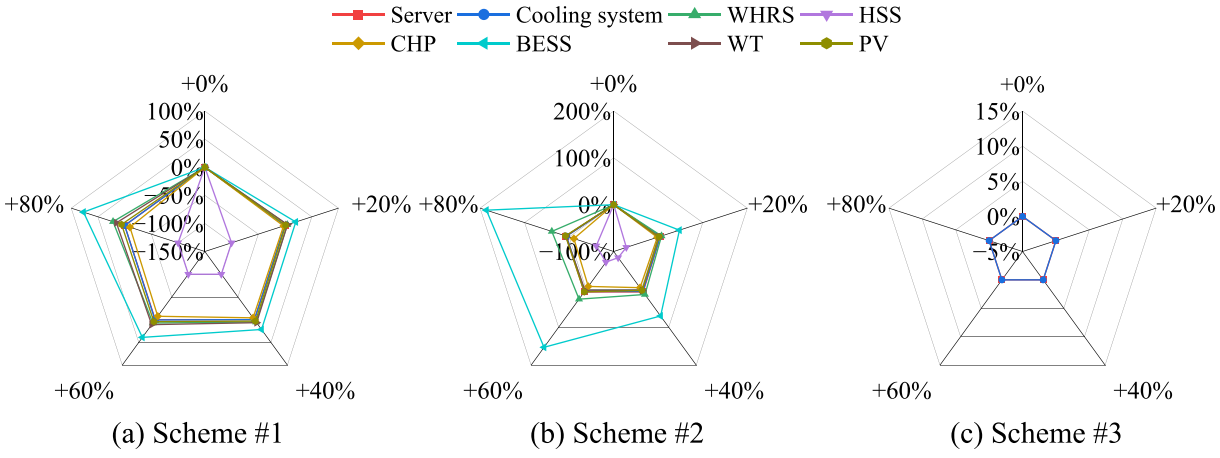


Fig. 16. Sensitivity of equipment capacity to fuel price.

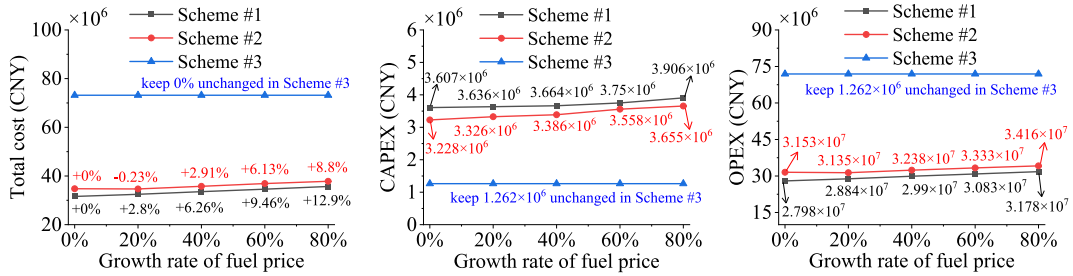


Fig. 17. Sensitivity of performance cost to fuel price.

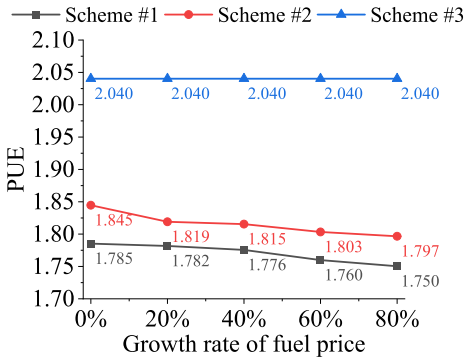


Fig. 18. Sensitivity of PUE to fuel price.

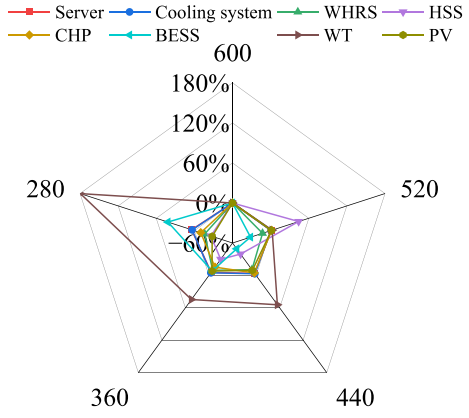


Fig. 19. Sensitivity of equipment capacity to scenario number in Scheme #1.

energy efficiency of the proposed solution.

4.6. Computational complexity analysis

The convergence curve of total cost across iterations is given in Fig. 22. The proposed heuristic bi-level solution approach achieves convergence after 12 iterations, with an average computation time of 3778.2 s for each iteration. Although the solution time before convergence is approximately 12.5 hours, it is acceptable for planning as it does not require real-time performance, as long as the optimal solution can be found.

5. Conclusions and future work

This paper focuses on the expansion planning method for upgrading DTCs and constructing CEHCSs. A data-driven scenario generation based expansion planning solution is proposed, which incorporates a DMOSG method to characterize the source-load uncertainties of CEHCSs. The main conclusions are summarized as follows:

1) In the proposed DMOSG, a 1dUN is adopted as the denoising network, and its latent output space is proven to be the logarithmic gradient of the target data distribution;
2) The DMOSG method is confirmed to be able to accurately learn the real data in terms of probability density distribution, cumulative distribution, order moment and temporal correlation behaviors;
3) The constructed CEHCS model considers the CPU temperature constraints of each server to ensure the operation safety. Concurrently, the PSO algorithm based bi-level scheme can cope with the nonlinearity of the CEHCS model by iteratively solving the expansion planning model. And the solution time is acceptable;
4) After upgrading the DTC with CEHCS, the combination of multiple energy equipment can significantly improve the cost-effectiveness and energy efficiency;
5) Compared with the planning scheme based on historical scenarios, the proposed data-driven scenario generation based planning scheme can more effectively consider the uncertainties of CEHCSs, enhance the system's ability to resist risks through more configured capacity and higher initial investment, thereby reducing long-term operating costs and further improving overall economic benefits.

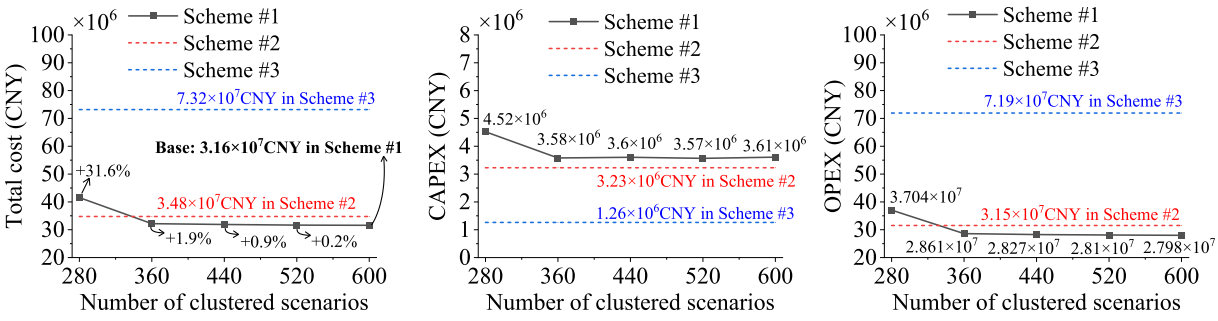


Fig. 20. Sensitivity of performance cost to scenario number.

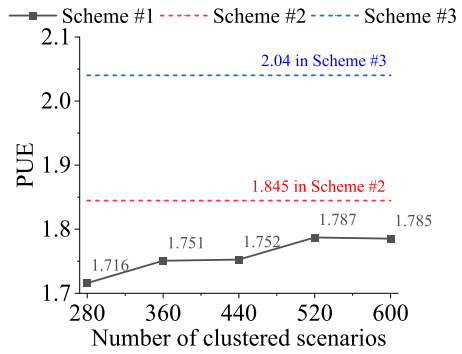


Fig. 21. Sensitivity of PUE to scenario number.

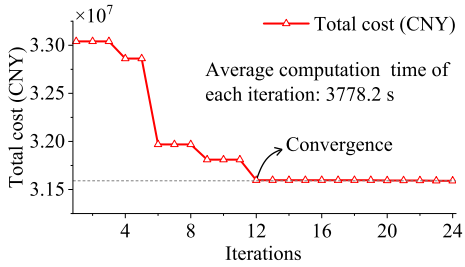


Fig. 22. Convergence curve of total cost across iterations in Scheme #1.

For future work, with the development of integrated energy systems, the energy supply to DTCs from different energy sources, e.g., hydrogen and gas, is worth further research exploitation. On the other hand, the controllable scenario generation technique also makes sense, which helps determine the planning scheme in special types of scenarios. Meanwhile, as the latent output space of DMOSG in this study is interpreted through theoretical derivation, a clear and intuitive method for visualizing this process has not yet been established. Therefore, future work will include validation through specific case studies to illustrate how interpretability is achieved in practice.

Besides, there are some modeling assumptions in this work. The direct-current power flow model is based on direct-current approximations, omitting voltage/reactive power considerations. The battery degradation, CHP maintenance cycles, and equipment outage risks are also not considered. These assumptions may result in a higher performance cost than expected for the proposed solution in real-world applications. Therefore, we plan to model the above components more realistically in future work. Further, the use of a small-scale IEEE 14-bus system and a single DTC limits the generalizability. Larger systems (e.g., IEEE 33 or 118) and multi-DTCs will be considered in future work to better demonstrate the framework's scalability.

CRediT authorship contribution statement

Lingfang Yang: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yujie Lin:** Writing – review & editing, Software, Investigation. **Mohammad Shahidehpour:** Writing – review & editing, Supervision, Investigation. **Yuanyi Chen:** Writing – review & editing. **Qiang Yang:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work is supported by the National Natural Science Foundation of China (52177119).

Appendix A. Derivation of the forward process in DMOSG

In the forward process of DMOSG , hyperparameters, i.e., α_s and β_s are used to adjust the amplitude of noises. The sum of these two parameters is 1, as shown in (A1).

$$\alpha_s = 1 - \beta_s \tag{A1}$$

where α_s and β_s are both positive; s is the index of diffusion steps, i.e., $s = 1, 2, \cdots, M$. During the diffusion process, β_s increases and α_s decreases monotonically.

The s-th and $(s-1)$-th diffusion step are shown in (A2) and (A3), respectively.

$$\mathbf{x}_s = \sqrt{\alpha_s}\mathbf{x}_{s-1} + \sqrt{1-\alpha_s}\boldsymbol{\varepsilon}_1 \tag{A2}$$

$$\mathbf{x}_{s-1} = \sqrt{\alpha_{s-1}}\mathbf{x}_{s-2} + \sqrt{1-\alpha_{s-1}}\boldsymbol{\varepsilon}_2 \tag{A3}$$

where $\mathbf{x}_{s-2}$, $\mathbf{x}_{s-1}$ and $\mathbf{x}_s$ are the data after adding noise; $\boldsymbol{\varepsilon}_1$ and $\boldsymbol{\varepsilon}_2$ are noises sampled from a standard Gaussian distribution, i.e., $\boldsymbol{\varepsilon}_1, \boldsymbol{\varepsilon}_2, \ldots \sim \mathcal{N}(0, \mathbf{I})$.

It can be seen that the forward diffusion process is a Markov process, and the result of each step (s) is related to the state of the previous step ($s-1$). To directly obtain the denoised data from the initial data, (A3) is substituted into (A2), resulting in (A4).

$$\begin{aligned} x_s &= \sqrt{\alpha_s}\left(\sqrt{\alpha_{s-1}}\,x_{s-2} + \sqrt{1-\alpha_{s-1}}\,\varepsilon_2\right) + \sqrt{1-\alpha_s}\varepsilon_1 \\ &= \sqrt{\alpha_s\alpha_{s-1}}x_{s-2} + \left(\sqrt{\alpha_s(1-\alpha_{s-1})}\,\varepsilon_2 + \sqrt{1-\alpha_s}\,\varepsilon_1\right) \\ &= \sqrt{\alpha_s\alpha_{s-1}}x_{s-2} + \sqrt{1-\alpha_s\alpha_{s-1}}\varepsilon_2 \end{aligned} \tag{A4}$$

Further, due to the fact that $\boldsymbol{\varepsilon}_s \sim \mathcal{N}(0, \mathbf{I})$, we can derive that $\mathrm{k}(\boldsymbol{\varepsilon}_s + \mathrm{b}) \sim \mathcal{N}(\mathrm{b}, \mathrm{k}^2\mathbf{I})$. Thus, (A5) and (A6) are obtained.

$$\sqrt{1-\alpha_s}\boldsymbol{\varepsilon}_1 \sim \mathcal{N}(0, 1-\alpha_s) \tag{A5}$$

$$\sqrt{\alpha_s(1-\alpha_{s-1})}\boldsymbol{\varepsilon}_2 \sim \mathcal{N}(0, \alpha_s(1-\alpha_{s-1})) \tag{A6}$$

Thus, the sum of $\sqrt{\alpha_s(1-\alpha_{s-1})}\boldsymbol{\varepsilon}_2$ and $\sqrt{1-\alpha_s}\boldsymbol{\varepsilon}_1$ follows the distribution $\mathcal{N}(0, 1-\alpha_s\alpha_{s-1})$, as shown in (A7).

$$\sqrt{1-\alpha_s\alpha_{s-1}}\varepsilon_2 \sim \mathcal{N}(0, 1-\alpha_s\alpha_{s-1}) \tag{A7}$$

As such, we can obtain the concise formulation of the forward process in DMOSG, as given in (A8).

$$\begin{aligned}\mathbf{x}_s &= \sqrt{(\alpha_s\alpha_{s-1}\alpha_{s-2}\alpha_{s-3}\cdots\alpha_1)}\mathbf{x}_0 + \sqrt{(1-\alpha_s\alpha_{s-1}\alpha_{s-2}\alpha_{s-3}\cdots\alpha_0)}\boldsymbol{\varepsilon}_s \\ &= \sqrt{\overline{\alpha}_s}x_0 + \sqrt{1-\overline{\alpha}_s}\boldsymbol{\varepsilon}_s\end{aligned} \tag{A8}$$

Appendix B. Derivation of the reverse process in DMOSG

In the forward process, $\mathbf{x}_s$ is obtained by gradually adding noise. This process requires the calculation of the forward conditional probability distribution, i.e., the prior distribution $q(\mathbf{x}_s|\mathbf{x}_{s-1}, \mathbf{x}_0)$. However, in the reverse process, the reverse conditional probability distribution needs to be calculated, i.e., the posterior distribution $q(\mathbf{x}_{s-1}|\mathbf{x}_s, \mathbf{x}_0)$, which represents calculating the probability distribution of $\mathbf{x}_{s-1}$ under the condition of knowing $\mathbf{x}_s$. The calculation of posterior probability distribution is much more complex than that of prior probability distribution, and requires gradual recursion through the Bayesian formula, as shown in (B1).

$$q(\mathbf{x}_{s-1}|\mathbf{x}_s, \mathbf{x}_0) = q(\mathbf{x}_s|\mathbf{x}_{s-1}, \mathbf{x}_0)\frac{q(\mathbf{x}_{s-1}|\mathbf{x}_0)}{q(\mathbf{x}_s|\mathbf{x}_0)} \tag{B1}$$

where $q(\mathbf{x}_s|\mathbf{x}_{s-1}, \mathbf{x}_0)$, $q(\mathbf{x}_{s-1}|\mathbf{x}_0)$ and $q(\mathbf{x}_s|\mathbf{x}_0)$ are three known Gaussian distributions. It can be derived from the forward process that the distribution of $q(\mathbf{x}_s|\mathbf{x}_{s-1}, \mathbf{x}_0)$, $q(\mathbf{x}_{s-1}|\mathbf{x}_0)$ and $q(\mathbf{x}_s|\mathbf{x}_0)$ follows (B2), (B3) and (B4), respectively.

$$q(\mathbf{x}_{s-1}|\mathbf{x}_0) = \sqrt{\overline{\alpha}_{s-1}}\mathbf{x}_0 + \sqrt{1-\overline{\alpha}_{s-1}}\boldsymbol{\varepsilon} \sim \mathcal{N}(\sqrt{\overline{\alpha}_{s-1}}\,\mathbf{x}_0, 1-\overline{\alpha}_{s-1}) \tag{B2}$$

$$q(\mathbf{x}_s|\mathbf{x}_0) = \sqrt{\overline{\alpha}_s}\mathbf{x}_0 + \sqrt{1-\overline{\alpha}_s}\boldsymbol{\varepsilon} \sim \mathcal{N}(\sqrt{\overline{\alpha}_s}\,\mathbf{x}_0, 1-\overline{\alpha}_s) \tag{B3}$$

$$q(\mathbf{x}_s|\mathbf{x}_{s-1}, \mathbf{x}_0) = \sqrt{\alpha_s}\mathbf{x}_{s-1} + \sqrt{1-\alpha_s}\boldsymbol{\varepsilon} \sim \mathcal{N}(\sqrt{\alpha_s}\,\mathbf{x}_{s-1},\ \ 1-\alpha_s) \tag{B4}$$

The formula for the Gaussian distribution is as (B5).

$$f(\boldsymbol{x}) = \frac{1}{\sigma\sqrt{2\pi}}\exp\left(-\frac{(\boldsymbol{x}\text{-}\mu)^2}{2\sigma^2}\right) \tag{B5}$$

Based on (B5), we can calculate the posterior probability distribution of the diffusion model, i.e., $q(\mathbf{x}_{s-1}|\mathbf{x}_s, \mathbf{x}_0)$, as shown in (B6).

$$\begin{aligned}q(x_{s-1}|x_s, x_0) &\propto \exp\left(-\frac{1}{2}\left(\frac{(x_s-\sqrt{\alpha_s}\,x_{s-1})^2}{\beta_s} + \frac{(x_{s-1}-\sqrt{\overline{\alpha}_{s-1}}\,x_0)^2}{1-\overline{\alpha}_{s-1}} - \frac{(x_s-\sqrt{\overline{\alpha}_s}\,x_0)^2}{1-\overline{\alpha}_s}\right)\right) \\ &= \exp\left(-\frac{1}{2}\left(\frac{x_s^2 - 2\sqrt{\alpha_s}\,x_s x_{s-1} + \alpha_s x_{s-1}^2}{\beta_s} + \frac{x_{s-1}^2 - 2\sqrt{\overline{\alpha}_{s-1}}\,x_0 x_{s-1} + \overline{\alpha}_{s-1}x_0^2}{1-\overline{\alpha}_{s-1}} - \frac{(x_s-\sqrt{\overline{\alpha}_s}\,x_0)^2}{1-\overline{\alpha}_s}\right)\right) \\ &= \exp\left(\begin{array}{c} -\frac{1}{2}\left(\left(\frac{\alpha_s}{\beta_s} + \frac{1}{1-\overline{\alpha}_{s-1}}\right)x_{s-1}^2 - \right. \\ \left.\left(\frac{2\sqrt{\alpha_s}}{\beta_s}x_s + \frac{2\sqrt{\overline{\alpha}_{s-1}}}{1-\overline{\alpha}_{s-1}}x_0\right)x_{s-1} + C(x_s, x_0)\right)\end{array}\right)\end{aligned} \tag{B6}$$

where the method of completing the square, i.e., (B7) is adopted.

$$\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) = \exp\left(-\frac{1}{2}\left(\frac{1}{\sigma^2}x^2 - \frac{2\mu}{\sigma^2}x + \frac{\mu^2}{\sigma^2}\right)\right) \tag{B7}$$

It can be derived from (B6) and (B7) that the standard deviation and mean of the distribution followed by the posterior distribution $q(\mathbf{x}_{s-1}|\mathbf{x}_s, \mathbf{x}_0)$ are as (B8) and (B9), respectively.

$$\frac{1}{\sigma^2} = \frac{\alpha_s}{\beta_s} + \frac{1}{1-\overline{\alpha}_{s-1}} = \frac{1-\overline{\alpha}_s}{\beta_s(1-\overline{\alpha}_{s-1})}, \text{i.e.,}\ \ \sigma^2 = \frac{\beta_s(1-\overline{\alpha}_{s-1})}{1-\overline{\alpha}_s} \tag{B8}$$

$$\widetilde{\boldsymbol{\mu}}_s(\mathbf{x}_s, \mathbf{x}_0) = \frac{\sqrt{\alpha_s}(1-\overline{\alpha}_{s-1})}{1-\overline{\alpha}_s}\mathbf{x}_s + \frac{\sqrt{\overline{\alpha}_{s-1}}\beta_s}{1-\overline{\alpha}_s}\mathbf{x}_0 \tag{B9}$$

As such, the variance of $\mathbf{x}_{s-1}$ is obtained, but the mean is unknown because of the unknown $\mathbf{x}_0$ in the reverse process. However, based on (A8) of the forward process, we can obtain (B10).

$$\mathbf{x}_0 = \frac{1}{\sqrt{\overline{\alpha}_s}}\left(\mathbf{x}_s - \sqrt{1-\overline{\alpha}_s}\,\boldsymbol{\varepsilon}_s\right) \tag{B10}$$

Then (B10) is substituted into (B9), and the mean of $\mathbf{x}_{s-1}$ can be obtained, as shown in (B11).

$$\widetilde{\mu}_s = \frac{1}{\sqrt{\alpha_s}}\left(\mathbf{x}_s - \frac{\beta_s}{\sqrt{1-\overline{\alpha}_s}}\boldsymbol{\varepsilon}_s\right) \tag{B11}$$

Thus, the variance and mean of the Gaussian distribution followed by $\mathbf{x}_{s-1}$ have been obtained. The variance is known, but $\boldsymbol{\varepsilon}_s$ in the mean cannot be

accurately obtained and can only be estimated. Therefore, a neural network model needs to be trained to fit ε_s. During training, the input parameters are $\mathbf{x}_s$ and s, and the output is iteratively compared with the label (noise added in the forward process) until convergence.

In summary, the posterior probability distribution of the diffusion model is shown in (B11).

$$q(x_{s-1}|x_s, x_0) = \mathcal{N}\left(x_{s-1}; \widetilde{\mu}_s(x_s, x_0), \widetilde{\beta}_s I\right), \tag{B11}$$

$$\text{where } \widetilde{\mu}_s(x_s, x_0) := \frac{\sqrt{\overline{\alpha}_{s-1}}\beta_s}{1-\overline{\alpha}_s}x_0 + \frac{\sqrt{\alpha_s}(1-\overline{\alpha}_{s-1})}{1-\overline{\alpha}_s}x_s \quad \text{and} \quad \widetilde{\beta}_s := \frac{1-\overline{\alpha}_{s-1}}{1-\overline{\alpha}_s}\beta_s$$

Appendix C. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.apenergy.2026.127369.

Data availability

Data will be made available on request.

References

[1] Wang F, Lv C. A data center expansion scheme considering net-zero carbon operation: Optimization of geographical location, on-site renewable utilization and green certificate purchase. Socioecon Plann Sci 2024;92:101789. https://doi.org/10.1016/j.seps.2023.101789.

[2] Yang T, Lee YC, Zomaya AY. Collective Energy-Efficiency Approach to Data Center Networks Planning. IEEE Transactions on Cloud Computing 2018;6(3):656–66. https://doi.org/10.1109/TCC.2015.2511732.

[3] Cáceres CR, Sandberg M, Sotoca A. Planning data center locations in Swedish municipalities. A comparative case study of Luleå and Stockholm. Cities 2024;150:105063. https://doi.org/10.1016/j.cities.2024.105063.

[4] Chen X, et al. A 10 MW class data center with ultra-dense high-efficiency energy distribution: Design and economic evaluation of superconducting DC busbar networks. Energy 2022;250:123820. https://doi.org/10.1016/j.energy.2022.123820.

[5] Zhou K, Fei Z, Lu X. Optimal Energy Management of Internet Data Center With Distributed Energy Resources. IEEE Trans On Cloud Comput 2023;11(3):2285–95. https://doi.org/10.1109/TCC.2022.3196655.

[6] Vafamehr A, Khodayar ME, Manshadi SD, Ahmad I, Lin J. A Framework for Expansion Planning of Data Centers in Electricity and Data Networks Under Uncertainty. IEEE Transactions on Smart Grid 2019;10(1):305–16. https://doi.org/10.1109/TSG.2017.2738444.

[7] Chen D, Ma Y, Wang L, Yao M. Spatio-temporal management of renewable energy consumption, carbon emissions, and cost in data centers. Sustainable Computing: Informatics and Systems 2024;41:100950. https://doi.org/10.1016/j.suscom.2023.100950.

[8] Yin X, Ye C, Ding Y, Song Y. Exploiting Internet Data Centers as Energy Prosumers in Integrated Electricity-Heat System. IEEE Trans Smart Grid 2023;14(1):167–82. https://doi.org/10.1109/TSG.2022.3197613.

[9] Li W, et al. Decentralized optimization for integrated electricity–heat systems with data center based energy hub considering communication packet loss. Appl Energy 2023;350:121586. https://doi.org/10.1016/j.apenergy.2023.121586.

[10] Yang D, et al. Prosumer data center system construction and synergistic optimization of computing power, electricity and heat from a global perspective. Thermal Science and Engineering Progress 2024;49:102469. https://doi.org/10.1016/j.tsep.2024.102469.

[11] Li W, Qian T, Zhang Y, Shen Y, Wu C, Tang W. Distributionally robust chance-constrained planning for regional integrated electricity–heat systems with data centers considering wind power uncertainty. Appl Energy 2023;336:120787. https://doi.org/10.1016/j.apenergy.2023.120787.

[12] Yin X, Ye C, Ding Y, Song Y, Wang L. Combined Heat and Power Dispatch Against Cold Waves Utilizing Responsive Internet Data Centers. IEEE Trans Sustain Energy 2024;15(2):819–34. https://doi.org/10.1109/TSTE.2023.3306529.

[13] Zou Y, Xu Y, Zhang C. A Risk-Averse Adaptive Stochastic Optimization Method for Transactive Energy Management of a Multi-Energy Microgrid. IEEE Trans. Sustain. Energy 2023;14(3):1599–611. https://doi.org/10.1109/TSTE.2023.3240184.

[14] Chen X, Dong W, Yang L, Yang Q. Scenario-based robust capacity planning of regional integrated energy systems considering carbon emissions. Renewable Energy 2023;207:359–75. https://doi.org/10.1016/j.renene.2023.03.030.

[15] Carmona R, Yang X. Joint Granular Model for Load, Solar and Wind Power Scenario Generation. IEEE Trans. Sustain. Energy 2024;15(1):674–86. https://doi.org/10.1109/TSTE.2023.3325202.

[16] Dong H, Wang L, Zhang X, Zeng M. A two-stage stochastic collaborative planning approach for data centers and distribution network incorporating demand response and multivariate uncertainties. Journal of Cleaner Production 2024;451:141482. https://doi.org/10.1016/j.jclepro.2024.141482.

[17] Chen H, Gao P, Liu K, Chen L, Huang W. Joint expansion planning for data centers and distribution networks based on conditional value-at-risk theory considering low carbon characteristics. Electric Power Systems Research 2024;229:110162. https://doi.org/10.1016/j.epsr.2024.110162.

[18] Yan L, Lue S, Yadi L, Li L. "A High Risk Operation Scenario Generation Method for High Renewable Energy Permeability Based on AND Operation," in 2023 5th International Conference on Power and Energy Technology (ICPET). 2023. p. 1554–60. https://doi.org/10.1109/ICPET59380.2023.10367199.

[19] Krishna AB, Abhyankar AR. An Efficient Data-Driven Conditional Joint Wind Power Scenario Generation for Day-Ahead Power System Operations Planning. IEEE Trans. Power Syst. 2024;39(2):3105–17. https://doi.org/10.1109/TPWRS.2023.3298004.

[20] Díaz G, Gómez-Aleixandre J, Coto J. Wind power scenario generation through state-space specifications for uncertainty analysis of wind power plants. Appl. Energy 2016;162:21–30. https://doi.org/10.1016/j.apenergy.2015.10.052.

[21] Tu Q, et al. Forecasting Scenario Generation for Multiple Wind Farms Considering Time-series Characteristics and Spatial-temporal Correlation. J. Mod Power Syst. Clean Energy 2021;9(4):837–48. https://doi.org/10.35833/MPCE.2020.000935.

[22] Wang Q, Zhang X, Xu D. Source-Load Scenario Generation Based on Weakly Supervised Adversarial Learning and Its Data-Driven Application in Energy Storage Capacity Sizing. IEEE Trans. Sustain. Energy 2023;14(4):1918–32. https://doi.org/10.1109/TSTE.2023.3258929.

[23] Qiao J, Pu T, Wang X. Renewable scenario generation using controllable generative adversarial networks with transparent latent space. CSEE Journal of Power and Energy Systems 2021;7(1):66–77. https://doi.org/10.17775/CSEEJPES.2020.00700.

[24] Chen X, Dong W, Yang Q. Robust optimal capacity planning of grid-connected microgrid considering energy management under multi-dimensional uncertainties. Appl Energy 2022;323:119642. https://doi.org/10.1016/j.apenergy.2022.119642.

[25] Wang Y, Liu Y, Yang Q. Operational Scenario Generation and Forecasting for Integrated Energy Systems. IEEE Trans. Ind. Informat. 2024;20(2):2920–31. https://doi.org/10.1109/TII.2023.3296875.

[26] Dong W, Chen X, Yang Q. Data-driven scenario generation of renewable energy production based on controllable generative adversarial networks with interpretability. Appl Energy 2022;308:118387. https://doi.org/10.1016/j.apenergy.2021.118387.

[27] Yuan R, Wang B, Sun Y, Song X, Watada J. Conditional Style-Based Generative Adversarial Networks for Renewable Scenario Generation. IEEE Trans. Power Syst. 2023;38(2):1281–96. https://doi.org/10.1109/TPWRS.2022.3170992.

[28] Ho J, Jain A, Abbeel P. Denoising Diffusion Probabilistic Models. 2020. https://doi.org/10.48550/arXiv.2006.11239. *arXiv*: arXiv:2006.11239.

[29] Song J, Meng C, Ermon S. Denoising Diffusion Implicit Models. 2020. https://doi.org/10.48550/arXiv.2010.02502.

[30] Dhariwal P, Nichol A. "Diffusion Models Beat GANs on Image Synthesis," in *Advances in Neural Information Processing Systems*, Curran Associates, Inc., 2021, pp. 8780–8794. Accessed: Aug. 07 [Online]. Available: https://proceedings.neurips.cc/paper/2021/hash/49ad23d1ec9fa4bd8d77d02681df5cfa-Abstract.html; 2024.

[31] Yang Z, Yan J, Hong Y, Yang J, Wang K, Li Y. "Scenario Generation of PV Power Based on DDPM," in 2023 IEEE 7th Conference on Energy Internet and Energy System Integration (EI2). 2023. p. 2062–6. https://doi.org/10.1109/EI259745.2023.10512827.

[32] Dong X, Mao Z, Sun Y, Xu X. Short-Term Wind Power Scenario Generation Based on Conditional Latent Diffusion Models. IEEE Trans. Sustain. Energy 2024;15(2):1074–85. https://doi.org/10.1109/TSTE.2023.3327497.

[33] Yang R, Li Y, Hu C, Hou FY. ExDiffusion: Classifier-Guidance Diffusion Model for Extreme Load Scenario Generation With Extreme Value Theory. IEEE Transactions on Smart Grid 2025;16(5):3887–903. https://doi.org/10.1109/TSG.2025.3584732.

[34] Li S, Xu C, Wei L, Li R, Ai X. "Scenario Generation of Renewable Energy Based on Improved Diffusion Model," in 2023 IEEE Sustainable Power and Energy Conference (iSPEC). 2023. p. 1–7. https://doi.org/10.1109/iSPEC58282.2023.10402981.

[35] Zhao W, Shao Z, Yang S, Lu X. A novel conditional diffusion model for joint source-load scenario generation considering both diversity and controllability. Appl. Energy 2025;377:124555. https://doi.org/10.1016/j.apenergy.2024.124555.

[36] Liu X. Research on collaborative scheduling of internet data center and regional integrated energy system based on electricity-heat-water coupling. Energy 2024; 292:130462. https://doi.org/10.1016/j.energy.2024.130462.
[37] Yang L, Chen X, Fang X, Yang Q. Optimal coordinated management of integrated electricity-heat-computation systems in geographically distributed data centers. Energy 2025;332:137232. https://doi.org/10.1016/j.energy.2025.137232.
[38] Zhang W, Yadav R, Tian Y-C, Tyagi SKS, Elgendy IA, Kaiwartya O. Two-Phase Industrial Manufacturing Service Management for Energy Efficiency of Data Centers. IEEE Trans. Ind. Informat. 2022;18(11):7525–36. https://doi.org/10.1109/TII.2022.3153508.
[39] Zhang S, Chatha KS. "Approximation algorithm for the temperature-aware scheduling problem," in *2007 IEEE/ACM International Conference on Computer-Aided Design*, Nov. 2007. https://doi.org/10.1109/ICCAD.2007.4397278. pp. 281–288.
[40] Dong W, Yang Q, Li W, Zomaya AY. Machine-Learning-Based Real-Time Economic Dispatch in Islanding Microgrids in a Cloud-Edge Computing Environment. IEEE Internet of Things Journal 2021;8(17):13703–11. https://doi.org/10.1109/JIOT.2021.3067951.
[41] "Multi-energy Systems Load from Arizona State University · IEEE Task Force on Data Sharing in Energy Systems." Accessed: July 09, 2025. [Online]. Available, https://ieee-pes-data-sharing.org/datasets/detail/2ff06e0f-af74-4c1c-8e7f-f7ec8a564ea0.
[42] "Parallel Workloads Archive: Logs." Accessed: Jan. 04, 2024. [Online]. Available, https://www.cs.huji.ac.il/labs/parallel/workload/logs.html.